

Multivariable Calculus

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1	Vectors and the Geometry of Space	2
1.1	Three-Dimensional Coordinate Systems	2
1.2	Introduction to Vectors	4
1.3	Products of Vectors	5
1.4	Projections of Vectors	8
1.5	Equations of Lines and Planes	9
1.6	Functions and Surfaces	17
1.7	Alternative Coordinate Systems	18
2	Vector Valued Functions	19
2.1	Vector Functions	19
2.2	Calculus with Vector Functions	20
2.3	Arc Length and Curvature	22
2.4	Applications of Vector Functions	30
2.5	Parametric Surfaces	32
3	Multivariate Limits and Derivatives	33
3.1	Graphs of Multivariate Functions	33
3.2	Limits and Continuity in Multiple Variables	33
3.3	Partial Derivatives	36
3.4	Tangent Planes & Approximations	39
3.5	Chain Rule	43
3.6	Gradients	47
3.7	Extreme Values	50
3.8	Lagrange Multipliers	54
4	Multivariate Integrals	58
4.1	Double Integrals	58
4.2	Iterative Integration	59
4.3	Double Integrals as Volume	60
4.4	Applications of Double Integrals	61
4.5	Triple Integrals	61
4.6	Interpretations of Triple Integrals	61
4.7	The Jacobian	61
5	Vector Calculus	61
5.1	Vector Fields	61
5.2	Line Integrals	63
5.3	Fundamental Theorem of Line Integrals	66
5.4	Green's Theorem	69
5.5	Divergence and Curl	72
5.6	Surface and Flux Integrals	75
5.7	Stokes' Theorem	80
5.8	Divergence Theorem	80
5.9	Generalizing Vector Calculus	83

1 Vectors and the Geometry of Space

1.1 Three-Dimensional Coordinate Systems

- A **plane** is two-dimensional because any point (x, y) on the plane can be represented using only two real numbers.
- A **space** is a three-dimensional collection of points, where each point is represented using three real numbers (x, y, z) .
 - We can set three orthogonal (perpendicular) lines that pass through a given origin $\mathcal{O} = (0, 0, 0)$. We can then call each of these lines the x -, y -, and z -axes respectively.
 - Typically, we imagine that the xy -plane (the plane which lies on the x - and y -axes) is at the bottom:
 - the x - and y -axes are “horizontal”.
 - the z -axis is “vertical”, with “upwards” indicating the positive z -direction.
 - The direction of the z -axis can be determined using the **right-hand rule**:
Extend your right hand in the direction of $+x$, then curl your fingers in the direction of $+y$. Your thumb will point in the direction of $+z$.
 - There are three **coordinate planes**: the xy -, xz -, and yz -planes.
 - The **projection** of a point P to a specific plane is the coordinates of P with the “other” coordinate set to 0.
ex: the projection of $(2, 5, 3)$ to the xz -axis is $(2, 0, 3)$. The y -coordinate is set to 0.
- n -**space** is the n -dimensional space represented by the collection of points represented by a collection of n real numbers.
- $\mathbb{R}^n$ is the set of all points in n -space.
 $\mathbb{R}^2$ is the set of all points in a plane, and $\mathbb{R}^3$ is the set of all points in 3-space.
 - If there is a bijection from coordinates to points in space (if a point in space can only ever be represented with one coordinate and vice versa), the coordinate system is **rectangular**.

Graphs in 3-space

- In 2-space, the set of points that satisfy an equation (e.g. one involving x and y) is called a **curve** in $\mathbb{R}^2$.
- Similarly, the set of points that satisfy equation in 3-space (e.g. one involving x , y , and z) is called a **surface** in $\mathbb{R}^3$.
- The following equations in $\mathbb{R}^3$ represent planes parallel to one of the coordinate planes:

$$x = c \quad y = c \quad z = c \quad \text{where } c \text{ is a constant.}$$

For example, the equation $z = 3$ represents all points $(x, y, 3)$, which is the xy -plane translated 3 units up (in the positive z -direction).

Distance between Two Points

The distance between two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ in 3-space is:

$$|PQ| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

Using the distance formula, we can derive the equation of the sphere of radius r to be the set of all points $P(x, y, z)$ a distance r from some center $\mathcal{C}(h, k, \ell)$:

$$\begin{aligned} |P\mathcal{C}| &= r \\ \sqrt{(x - h)^2 + (y - k)^2 + (z - \ell)^2} &= r \\ (x - h)^2 + (y - k)^2 + (z - \ell)^2 &= r^2. \end{aligned}$$

Example: Finding the center and radius of a sphere

Find the center and radius of the sphere $x^2 + y^2 + z^2 + 4x - 6y + 2z + 6 = 0$.

Rewrite into standard form by completing the squares:

$$\begin{aligned} (x^2 + 4x) + (y^2 - 6y) + (z^2 + 2z) + 6 &= 0 \\ (x^2 + 4x + 4) + (y^2 - 6y + 9) + (z^2 + 2z + 1) + 6 &= 4 + 9 + 1 \\ (x + 2)^2 + (y - 3)^2 + (z + 1)^2 &= 4 + 9 + 1 - 6 \\ (x + 2)^2 + (y - 3)^2 + (z + 1)^2 &= 8 = \underbrace{\sqrt{8}^2}_r \end{aligned}$$

Thus, the center is $(-2, 3, -1)$ and the sphere has radius $r = \sqrt{8} = 2\sqrt{2}$.

1.2 Introduction to Vectors

- Vectors are either denoted with arrows or they are bolded.
 - If v is a vector, then we can clarify so by writing $\vec{v}$ or $\mathbf{v}$.
- If the points A and B are defined, the vector pointing from A to B can be written $\overrightarrow{AB}$.
- A vector in n -dimensions is composed of n *components*. Each vector component represents a certain direction in space.
- A vector can be written in component form, either as a 1-D matrix (a “column vector”) or by using angle brackets:

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \langle v_1, v_2, \dots, v_n \rangle.$$

- The **magnitude** of a vector is the distance from head to tail. The magnitude of a vector $\vec{v}$ is denoted $|\vec{v}|$ or $\|\vec{v}\|$. If $\vec{v} = \langle v_1, v_2, \dots, v_n \rangle$, then its magnitude $|\vec{v}| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$.
- A **unit vector** is a vector with magnitude 1. If $\vec{v}$ is a well-defined unit vector, then it is typically denoted $\hat{v}$ (“*v-hat*”), and $|\hat{v}| = 1$.
 - The three *basis vectors*, or fundamental directional unit vectors, are $\hat{i}$, $\hat{j}$, and $\hat{k}$. It holds that $\langle x, y, z \rangle = x\hat{i} + y\hat{j} + z\hat{k}$.
 - The unit vector $\hat{v}$ of a vector $\vec{v}$ points in the same direction as $\vec{v}$, and $\hat{v} = \frac{\vec{v}}{|\vec{v}|}$.
- The **zero vector** is the only vector with a magnitude of 0, denoted $\vec{0}$.
- A vector only describes a displacement, not an absolute position. A vector has no way of describing where it starts and ends without some reference point.

Combining Vectors

- The sum of two vectors $\vec{a} = \langle a_1, a_2, \dots, a_n \rangle$ and $\vec{b} = \langle b_1, b_2, \dots, b_n \rangle$ is:

$$\vec{a} + \vec{b} = \langle a_1, a_2, \dots, a_n \rangle + \langle b_1, b_2, \dots, b_n \rangle = \langle a_1 + b_1, a_2 + b_2, \dots, a_n + b_n \rangle.$$

In other words, it is just adding each corresponding component together.

- The difference of two vectors works the same way.
- It holds that $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$ (triangle law) and $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ (parallelogram law), each named for how the property can be visualized on a graph.

Scalar Products

Multiplying a vector by a scalar (or vice-versa) is the equivalent of multiplying each component by the scalar:

$$k\vec{v} = k\langle v_1, v_2, \dots, v_n \rangle = \langle kv_1, kv_2, \dots, kv_n \rangle \quad \text{where } k \text{ is any scalar.}$$

It also holds that $|k\vec{v}| = k|\vec{v}|$.

1.3 Products of Vectors

There are two fundamental ways to “combine” two vectors: by using a **dot product** and by using a **cross product**.

The Dot Product

The dot product of two vectors $\vec{a}$ and $\vec{b}$, denoted $\vec{a} \cdot \vec{b}$, interprets the product as both how much the two vectors point in the same direction as well as the scalar product of their magnitudes.

- Geometrically, if θ is the angle between the vectors $\vec{a}$ and $\vec{b}$, then $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta$.
 - Thus, if $\vec{a}$ is orthogonal to $\vec{b}$, then $\vec{a} \cdot \vec{b} = 0$ because $\cos 90^\circ = 0$.
 - Additionally, if $\vec{a}$ points in the same direction as $\vec{b}$, then $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|$ because $\cos 0^\circ = 1$.
- For example, work can be defined as the dot product of force $\mathbf{F}$ and displacement $\mathbf{r}$, since only the component of force directed along the displacement should be considered ($W = \mathbf{F} \cdot \mathbf{r}$).

Derivation: Dot Product using Components

The first principles of the dot product is actually the geometric definition of the dot product:

$$\vec{a} \cdot \vec{b} = ab \cos \theta$$

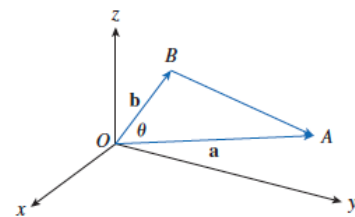
...where $a = |\vec{a}|$, $b = |\vec{b}|$, and θ is the angle between $\vec{a}$ and $\vec{b}$ ($0 \leq \theta \leq \pi$).

It turns out that a , b , and $|\vec{a} - \vec{b}|$ are vectors that make up the side lengths of a triangle (see right figure).

Let $\vec{a} = \langle a_1, a_2, \dots, a_n \rangle$ and $\vec{b} = \langle b_1, b_2, \dots, b_n \rangle$.

By the law of cosines:

$$\begin{aligned} |\vec{a} - \vec{b}|^2 &= a^2 + b^2 - 2ab \cos \theta \\ &= a^2 + b^2 - 2(\vec{a} \cdot \vec{b}). \end{aligned}$$



We can then isolate the dot product:

$$\begin{aligned} \vec{a} \cdot \vec{b} &= \frac{1}{2} \left(a^2 + b^2 - |\vec{a} - \vec{b}|^2 \right) \\ &= \frac{1}{2} \left(a_1^2 + a_2^2 + \dots + a_n^2 \right. \\ &\quad \left. + b_1^2 + b_2^2 + \dots + b_n^2 \right. \\ &\quad \left. - (a_1 - b_1)^2 - (a_2 - b_2)^2 - \dots - (a_n - b_n)^2 \right) \\ &= \frac{1}{2} \left(a_1^2 + a_2^2 + \dots + a_n^2 \right. \\ &\quad \left. + b_1^2 + b_2^2 + \dots + b_n^2 \right. \\ &\quad \left. - (a_1^2 - 2a_1b_1 + b_1^2) - (a_2^2 - 2a_2b_2 + b_2^2) - \dots - (a_n^2 - 2a_nb_n + b_n^2) \right) \\ &= \frac{1}{2} (2a_1b_1 + 2a_2b_2 + \dots + 2a_nb_n) = a_1b_1 + a_2b_2 + \dots + a_nb_n. \end{aligned}$$

Dot Product of Two Vectors (Component Form)

Let $\vec{a} = \langle a_1, a_2, \dots, a_n \rangle$ and $\vec{b} = \langle b_1, b_2, \dots, b_n \rangle$ be vectors in n -dimensions. Then:

$$\vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i = a_1 b_1 + a_2 b_2 + \dots + a_n b_n.$$

The Cross Product

Cross Product of Two Vectors in 3D

Let $\vec{a} = \langle a_x, a_y, a_z \rangle$ and $\vec{b} = \langle b_x, b_y, b_z \rangle$. Then:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = \begin{pmatrix} a_y b_z - a_z b_y \\ a_z b_x - a_x b_z \\ a_x b_y - a_y b_x \end{pmatrix}$$

- Unlike dot products which output scalar values, cross products output vectors.
 - The *right hand rule* can be used to determine the direction of the resulting vector.
If the vectors $\vec{a}$ and $\vec{b}$ lie on a plane P , then $\vec{a} \times \vec{b}$ is orthogonal to P .
 - The unit vector pointing in the direction of the resulting vector is typically denoted $\hat{n}$. This is called the *normal vector* or the *unit normal*.
- Geometrically, if θ is the angle between the vectors $\vec{a}$ and $\vec{b}$, then $\vec{a} \times \vec{b} = (|\vec{a}||\vec{b}| \sin \theta) \hat{n}$.
 - Thus, if $\vec{a}$ points in the same direction as $\vec{b}$, then $\vec{a} \times \vec{b} = 0$ because $\sin 0^\circ = 0$.
This makes sense because $\vec{a}$ and $\vec{b}$ no longer form a plane (they are not *coplanar*).
 - Additionally, if $\vec{a}$ is orthogonal to $\vec{b}$, then $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|$ because $\sin 90^\circ = 1$.
- **Cross products are not commutative:** $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$
(if we flip the order, the cross product will point in the opposite direction)
- **Cross products are not associative:** $(\vec{a} \times \vec{b}) \times \vec{c} \neq \vec{a} \times (\vec{b} \times \vec{c})$.
- Cross products are an interpretation of the area of the parallelogram formed by the vectors.
 $|\vec{a} \times \vec{b}|$ is the area of the parallelogram bounded by $\vec{a}$ and $\vec{b}$!

Example: Area of a Triangle given Three Points

Find the area of the triangle ABC with vertices $A(2, 1, 2)$, $B(3, 5, 4)$, and $C(1, 6, 7)$.

Although we can try computing the lengths of each side of the triangle and apply triangle formulas to solve this problem, we can just as well use a cross product.

The area of the parallelogram with adjacent sides AB and BC is $|\overrightarrow{AB} \times \overrightarrow{BC}|$.

Find $\overrightarrow{AB} \times \overrightarrow{BC}$ and compute its magnitude:

$$\begin{aligned}\overrightarrow{AB} \times \overrightarrow{BC} &= \langle 3 - 2, 5 - 1, 4 - 2 \rangle \times \langle 1 - 3, 6 - 5, 7 - 4 \rangle \\ &= \langle 1, 4, 2 \rangle \times \langle -2, 1, 3 \rangle \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 4 & 2 \\ -2 & 1 & 3 \end{vmatrix} = \langle 10, -7, 9 \rangle.\end{aligned}$$

$$\text{Area}_{\text{parallelogram}} = |\overrightarrow{AB} \times \overrightarrow{BC}| = \sqrt{10^2 + (-7)^2 + 9^2} = \sqrt{230}.$$

The area of the triangle is half the area of the parallelogram:

$$\text{Area}_{\triangle ABC} = \frac{1}{2} \text{Area}_{\text{parallelogram}} = \frac{\sqrt{230}}{2}.$$

Cross Products of Basis Vectors

You should use the right-hand rule to memorize the following cross products:

$$\begin{array}{ll}\hat{i} \times \hat{j} = \hat{k} & \hat{j} \times \hat{i} = -\hat{k} \\ \hat{j} \times \hat{k} = \hat{i} & \hat{k} \times \hat{j} = -\hat{i} \\ \hat{k} \times \hat{i} = \hat{j} & \hat{i} \times \hat{k} = -\hat{j}\end{array}$$

Triple Products

- The **scalar triple product** of three vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ is $\vec{a} \cdot (\vec{b} \times \vec{c})$. The triple product is relevant since it is the volume of the parallelepiped (parallelogram prism) formed by the three vectors.

If the volume of the parallelepiped is 0, then the triple product is 0 and our vectors lie on the same plane (the parallelepiped is flat). In other words, $\vec{a}$, $\vec{b}$, and $\vec{c}$ are **coplanar** if $\vec{a} \cdot (\vec{b} \times \vec{c}) = 0$.

Conveniently, it also holds that $\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$.

- The **vector triple product** of three vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ is $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$.

1.4 Projections of Vectors

The **projection** of a vector $\vec{a}$ along another vector $\vec{b}$ is the component of $\vec{a}$ which points in the same direction as $\vec{b}$. There are two types of projections:

- The **scalar projection** is simply the magnitude of such component. We can derive it from the definition of the dot product:

Derivation: Scalar Projection

1. Let $a = |\vec{a}|$ and $b = |\vec{b}|$.
2. Since $\vec{a} \cdot \vec{b} = ab \cos \theta$, where θ is the angle between $\vec{a}$ and $\vec{b}$, $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{ab}$.
3. Using trigonometry, $\cos \theta = \frac{x}{a}$, where x is the scalar component of $\vec{a}$ along $\vec{b}$.
4. $\therefore \frac{x}{a} = \frac{\vec{a} \cdot \vec{b}}{ab} \implies x = \frac{\vec{a} \cdot \vec{b}}{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$.

- The scalar projection is often represented using the syntax $\text{comp}_{\vec{b}} \vec{a}$, where $\vec{b}$, the subscript, is what we are projecting *onto*.

Scalar Projection

The **scalar projection** of $\vec{a}$ along (“onto”) $\vec{b}$, $\text{comp}_{\vec{b}} \vec{a}$, is the signed magnitude of the component of $\vec{a}$ which is parallel to $\vec{b}$:

$$\text{comp}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$$

- The **vector projection** of $\vec{a}$ along $\vec{b}$, $\text{proj}_{\vec{b}} \vec{a}$, is the vector component of $\vec{a}$ pointing in the same direction as $\vec{b}$. It is simply the scalar projection times the unit vector for $\vec{b}$:

$$\text{proj}_{\vec{b}} \vec{a} = \text{comp}_{\vec{b}} \vec{a} \cdot \hat{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} \cdot \frac{\vec{b}}{|\vec{b}|} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}.$$

Vector Projection

The **vector projection** of $\vec{a}$ along (“onto”) $\vec{b}$, $\text{proj}_{\vec{b}} \vec{a}$, is the vector component of $\vec{a}$ pointing in the same direction as $\vec{b}$:

$$\text{proj}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}.$$

- **proj** is often bolded because the operation results in a vector.
- The result of the projection specified by **proj** is in the direction of the *subscript*.

1.5 Equations of Lines and Planes

Point-Slope Equations of Lines

We know that in the xy -plane, we can define a line using its *point-slope form*, which specifies a point through which the line passes as well as the slope of the line.

A line which passes through the point (x_0, y_0) with slope m can be described using the equation $y = m(x - x_0) + y_0$. This is derived from applying translations to $y = mx$.

In a similar way, we can define a line in 3-space using a point through which the line passes as well as a *vector* which lies parallel to the line.

(In a way, our line has “multiple” slopes, which are all contained in the vector.)

Derivation: Parametric Equations for a Line in 3-space

1. Let the line L in 3-space pass through the point (x_0, y_0, z_0) . We can define the point using a **position vector**, $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$.

Now, assume the same line L points in the same direction as some vector $\vec{v} = \langle a, b, c \rangle$. We call a , b , and c the **direction numbers** for L .

Finally, let (x, y, z) be an arbitrary point on L , and $\vec{r} = \langle x, y, z \rangle$ is its position vector.

2. We can arrive at $\vec{r}$ by stepping in the direction of $\vec{v}$ starting from $\vec{r}_0$:

$$\vec{r} = \vec{r}_0 + t\vec{v}.$$

We introduce the parameter $t \in \mathbb{R}$ as an arbitrary scale factor of $\vec{v}$, since $\vec{v}$ does not specify the correct magnitude, only direction. Writing out the vector components:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} + t \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} x_0 + at \\ y_0 + bt \\ z_0 + ct \end{pmatrix}$$

3. This is a set of **parametric equations** which define the line L :

$$x(t) = x_0 + at \quad y(t) = y_0 + bt \quad z(t) = z_0 + ct$$

Point-Slope Equation of a Line

Let L be the line which passes through the point (x_0, y_0, z_0) and lies parallel to some vector $\vec{v} = \langle a, b, c \rangle$. Then a point (x, y, z) lies on L iff:

$$x(t) = x_0 + at \quad y(t) = y_0 + bt \quad z(t) = z_0 + ct \quad \text{where } t \in \mathbb{R}.$$

If $\vec{r} = \langle x, y, z \rangle$ and $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$, then L can also be represented by this vector equation:

$$\vec{r} = \vec{r}_0 + t\vec{v}.$$

Note

There can be many different sets of parametric equations which define the same lines in 3-space. We can change the point or choose a completely different direction vector.

Example: Equations of a Line from Point and Direction

Write a set of parametric equations which define the line that passes through the point $(5, 1, 3)$ and points in the same direction as $\hat{i} + 4\hat{j} - 2\hat{k}$.

Write each vector in component form:

$$\begin{aligned}\vec{r} &= \langle x, y, z \rangle \\ \vec{r}_0 &= \langle x_0, y_0, z_0 \rangle = \langle 5, 1, 3 \rangle \\ \vec{v} &= \langle a, b, c \rangle = \langle 1, 4, -2 \rangle\end{aligned}$$

From $\vec{r} = \vec{r}_0 + t\vec{v}$, we get the following parametric equations:

$$x(t) = 5 + t \qquad y(t) = 1 + 4t \qquad z(t) = 3 - 2t$$

Example: Equations of a Line from two Points

Write a set of parametric equations which define the line L that passes through the points $(6, 2, 7)$ and $(1, 4, 5)$. At what point does L intersect the xy -plane?

This is a simple extension from the content above, since the direction vector of a line passing through two points represented by position vectors $\vec{a}$ and $\vec{b}$ is just $\vec{a} - \vec{b}$ or $\vec{b} - \vec{a}$.

Let's choose $(1, 4, 5)$ as our given point, $\vec{r}_0 = \langle 1, 4, 5 \rangle$.

The direction vector can be computed as $\vec{v} = \langle 6, 2, 7 \rangle - \langle 1, 4, 5 \rangle = \langle 5, -2, 2 \rangle$.

We now have everything to write the parametric equations:

$$x(t) = 1 + 5t \qquad y(t) = 4 - 2t \qquad z(t) = 5 + 2t$$

The line will intersect the xy -plane when $z = 0$:

$$z(t) = 5 + 2t = 0 \implies t = -\frac{5}{2}.$$

Plug in $t = -\frac{5}{2}$ to every other equation in our parametric to find the point of intersection:

$$\begin{aligned}x\left(-\frac{5}{2}\right) &= 1 + 5\left(-\frac{5}{2}\right) = -\frac{23}{2} \\ y\left(-\frac{5}{2}\right) &= 4 - 2\left(-\frac{5}{2}\right) = 9\end{aligned}$$

(we already know $z = 0$)

Thus, the point that lies on L and intersects the xy -plane is $\left(-\frac{23}{2}, 9, 0\right)$.

Symmetric Equations of Lines

For each parametric equation, we can isolate t :

$$t = \frac{x - x_0}{a} \qquad t = \frac{y - y_0}{b} \qquad t = \frac{z - z_0}{c}$$

This creates a three-way equality statement, known as the **symmetric equations** of the line:

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c} \quad \text{where } a, b, c \neq 0$$

If any x , y , or z is 0, then we cannot include the respective expression in our equation. For example, if $a = 0$, our symmetric equations become:

$$x = x_0, \quad \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

This means the line lies completely on $x = x_0$.

Equations of Line Segments

The parametric equations of line *segments* are constrained by the endpoints. Specifically, we can turn a set of parametric equations for a line into a line segment by constraining t to an interval.

For example, the following describes a line segment:

$$x(t) = 5 + t \quad y(t) = 1 + 4t \quad z(t) = 3 - 2t \quad 0 \leq t \leq 1$$

This line segment has endpoints $\vec{r}(0) = (5, 1, 3)$ and $\vec{r}(1) = (6, 5, 1)$.

Derivation: General Equation of a Line Segment

When finding the equations for a line segment given two endpoints and a constraint for t , we must choose the correct direction vector such that the constraints are met.

If given a direction vector $\vec{v}$, we must use $k\vec{v}$ where k is some unknown scale of $\vec{v}$ that will allow our final equations to meet the constraint.

Let a line segment L have endpoints $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$ and $\vec{r}_n = \langle x_n, y_n, z_n \rangle$. There exists a vector $\vec{v} = \langle a, b, c \rangle$ that is parallel to L . Find a set of parametric equations in t which represent the line segment, where $0 \leq t \leq n$.

$\vec{v} \parallel L \implies k\vec{v} \parallel L \implies \langle ka, kb, kc \rangle \parallel L$. Our vector equation is:

$$\vec{r}(t) = \vec{r}_0 + tk\vec{v} \quad 0 \leq t \leq n.$$

which when written as a parametric is:

$$x(t) = x_0 + akt \quad y(t) = y_0 + bkt \quad z(t) = z_0 + ckt \quad 0 \leq t \leq n.$$

Solve for k when $t = n$ (only one of the below statements are needed):

$$k = \frac{x_n - x_0}{an} = \frac{y_n - y_0}{bn} = \frac{z_n - z_0}{cn}$$

Thus, the equations become:

$$x(t) = x_0 + \frac{x_n - x_0}{n}t \quad y(t) = y_0 + \frac{y_n - y_0}{n}t \quad z(t) = z_0 + \frac{z_n - z_0}{n}t \quad 0 \leq t \leq n.$$

In vector form:

$$\vec{r}(t) = \vec{r}_0 + \frac{\vec{r}_n - \vec{r}_0}{n}t = \vec{r}_0 + \frac{1}{n}\Delta\vec{r}t.$$

Equations of Planes

A plane can also be described using a point which lies on the plane and a direction vector. However, a vector parallel to a plane is not sufficient to describe its direction (there are an infinite amount of planes that would satisfy such condition). Instead, the direction vector is a vector perpendicular to the plane, called the **normal vector** $\vec{n}$.

Derivation: Scalar Equation of a Plane

1. Let the vector $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$ represent a point that lies on P .
Let the vector $\vec{n} = \langle a, b, c \rangle$ be *orthogonal* to P , indicating its direction.
Let the point represented by $\vec{r} = \langle x, y, z \rangle$ be an arbitrary point which lies on P .
2. The vector pointing from $\vec{r}_0$ to $\vec{r}$, $\vec{r} - \vec{r}_0$, must lie on P because both $\vec{r}_0$ and $\vec{r}$ lie on P .
Since $\vec{n}$ is orthogonal to every vector which completely lies on P , $\vec{n}$ is orthogonal to $\vec{r} - \vec{r}_0$.

The dot product of orthogonal vectors is always 0:

$$\begin{aligned}\vec{n} \cdot (\vec{r} - \vec{r}_0) &= 0 \quad (\text{vector equation}) \\ \langle a, b, c \rangle \cdot \langle x - x_0, y - y_0, z - z_0 \rangle &= 0 \\ a(x - x_0) + b(y - y_0) + c(z - z_0) &= 0.\end{aligned}$$

This is the **scalar equation** of the plane P .

Equation of a Plane

If a plane P in 3-space contains the point represented by position vector $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$ and the vector $\vec{n} = \langle a, b, c \rangle$ is orthogonal to P , then an arbitrary point represented by position vector $\vec{r} = \langle x, y, z \rangle$ lies on P iff:

$$\begin{aligned}\vec{n} \cdot (\vec{r} - \vec{r}_0) &= 0 \quad (\text{vector equation}) \\ \implies a(x - x_0) + b(y - y_0) + c(z - z_0) &= 0 \quad (\text{scalar equation}).\end{aligned}$$

Example: Equation of a Plane given Point and Normal Vector

The plane P that passes through the point $(2, 4, -1)$. If the vector $2\hat{i} + 3\hat{j} + 4\hat{k}$ is orthogonal to P , find an equation for P .

Using $\vec{r}_0 = \langle 2, 4, -1 \rangle$ and $\vec{n} = \langle 2, 3, 4 \rangle$, we get:

$$2(x - 2) + 3(y - 4) + 4(z + 1) = 0.$$

This can be simplified into standard form ($ax + by + cz = d$):

$$\begin{aligned}2x - 4 + 3y - 12 + 4z + 4 &= 0 \\ 2x + 3y + 4z &= 12.\end{aligned}$$

Example: Equation of a Plane given Three Points

The plane P passes through the following set of points: $\{(1, 3, 2), (3, -1, 6), (5, 2, 0)\}$. Find an equation for P .

Any three points can be used to describe a plane, as long as they are not *colinear*. We can create two vectors out of the three points which are parallel to P :

$$\vec{a} = \langle 3, -1, 6 \rangle - \langle 1, 3, 2 \rangle = \langle 2, -4, 4 \rangle$$

$$\vec{b} = \langle 5, 2, 0 \rangle - \langle 3, -1, 6 \rangle = \langle 2, 3, -6 \rangle$$

The cross product of these two vectors will be orthogonal to P , and we can use it as our normal vector:

$$\begin{aligned}\vec{n} = \vec{a} \times \vec{b} &= \langle 2, -4, 4 \rangle \times \langle 2, 3, -6 \rangle = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -4 & 4 \\ 2 & 3 & -6 \end{vmatrix} \\ &= \begin{vmatrix} -4 & 4 \\ 3 & -6 \end{vmatrix} \hat{i} - \begin{vmatrix} 2 & 4 \\ 2 & -6 \end{vmatrix} \hat{j} + \begin{vmatrix} 2 & -4 \\ 2 & 3 \end{vmatrix} \hat{k} \\ &= 12\hat{i} + 20\hat{j} + 14\hat{k} = \langle 12, 20, 14 \rangle.\end{aligned}$$

Finally, we can select one of the points again to write the equation of P .

Selecting $\vec{r}_0 = \langle 1, 3, 2 \rangle$, we get:

$$12(x - 1) + 20(y - 3) + 14(z - 2) = 0$$

$$12x - 12 + 20y - 60 + 14z - 28 = 0$$

$$12x + 20y + 14z = 100$$

$$6x + 10y + 7z = 50.$$

Intersections of Planes

- The angle of intersection between two planes is the angle between their normal vectors. If the normal vectors of two intersecting planes are $\vec{n}_1$ and $\vec{n}_2$, then the angle θ between them is:

$$\theta = \arccos\left(\frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1||\vec{n}_2|}\right).$$

- Two planes can either intersect nowhere, everywhere, or along a single line, called the *line of intersection* between the two planes.
 - If two planes intersect nowhere or everywhere, they are parallel. Two planes are parallel if their normal vectors are also parallel.
- The equations of two intersecting planes create a system of two equations with three unknowns. This means one of our unknowns are free to vary, which is why the set of solutions to the system form a line.
 - The set of parametric equations for the line of intersection can be obtained by parameterizing the free variable.

Example: Line of Intersection of Two Planes

L is the line of intersection between the planes $2x + 3y + 5z = 1$ and $-x + 2y + 3z = 2$. Find a set of parametric equations that represent L .

This is a system of two equations with three unknowns:

$$\begin{aligned}2x + 3y + 5z &= 1 \\ -x + 2y + 3z &= 2\end{aligned}$$

Choose to eliminate two variables. First let's eliminate x :

$$\begin{array}{rcl}2x + 3y + 5z &= & 1 \\ -2x + 4y + 6z &= & 4 \\ \hline 7y + 11z &= & 5\end{array} \implies y = \frac{5}{7} - \frac{11}{7}z.$$

Then, substitute y in terms of z into one of the original equations and solve for x :

$$\begin{aligned}-x + 2y + 3z &= 2 \\ -x + 2\left(\frac{5}{7} - \frac{11}{7}z\right) + 3z &= 2 \\ \frac{10}{7} - \frac{22}{7}z + 3z &= 2 + x \\ -\frac{4}{7} - \frac{1}{7}z &= x.\end{aligned}$$

Now, we have two functions of z : $x(z)$ and $y(z)$. If we let $z = t$, then we have now have a set of three parametric equations for L :

$$x(t) = -\frac{4}{7} - \frac{1}{7}t \quad y(t) = \frac{5}{7} - \frac{11}{7}t \quad z(t) = t.$$

Or, if we let $z = 7t$, we can eliminate a few fractions from our equations:

$$x(t) = -\frac{4}{7} - t \quad y(t) = \frac{5}{7} - 11t \quad z(t) = 7t.$$

Distances Between Planes

The distance between a point and a plane can be calculated using a scalar projection. If P is a point on a plane S , then a vector pointing from Q to P is $\overrightarrow{PQ}$. The shortest distance from Q to S is the scalar component of $\overrightarrow{PQ}$ that is parallel to the vector normal to P .

Distance from a Point to a Plane

Let P and Q be points in $\mathbb{R}^3$, and S is a plane containing P with normal vector $\vec{n}$. The distance from Q to the plane S is given by:

$$|QS| = |\text{comp}_{\vec{n}} \overrightarrow{PQ}| = \frac{|\vec{n} \cdot \overrightarrow{PQ}|}{|\vec{n}|}.$$

Extending this, we can find the distance from a plane A to another plane B ($A \parallel B$) by choosing an arbitrary point on one of the planes and using the above formula.

Skew Lines

Two lines are **skew** if they are not parallel and they do not intersect. Thus, skew lines must lie on parallel but separate *planes*. The distance between the two lines is equal to the distance between the respective parallel planes they lie on.

Example: Verify Two Lines are Skew

Show that the following lines A and B are *skew*:

$$\begin{array}{lll} A : & x = 1 + t & y = -2 + 3t & z = 4 - t \\ B : & x = 2s & y = 3 + s & z = -3 + 4s \end{array}$$

1. First, we can verify that the two lines are not parallel to each other. The direction vectors of A and B are $\langle 1, 3, -1 \rangle$ and $\langle 2, 1, 4 \rangle$ respectively, which are not scalar multiples of each other.
2. Next, we want to show that a point of intersection (x, y, z) does not exist. We can write a system of equations in t and s by equating each component:

$$\begin{aligned} 1 + t &= 2s \\ -2 + 3t &= 3 + s \\ 4 - t &= -3 + 4s. \end{aligned}$$

We can solve for t and s using two of the equations and verify that there is a contradiction when those values of t and s are plugged into the other.

Using the first two equations, we get $t = \frac{11}{5}$ and $s = \frac{8}{5}$:

$$\begin{aligned} 1 + t &= 2s \implies t = 2s - 1 \\ -2 + 3(2s - 1) &= 3 + s \implies s = \frac{8}{5}, t = 2\left(\frac{8}{5}\right) - 1 = \frac{11}{5} \end{aligned}$$

Those values of t and s give us a contradiction in the third equation:

$$\begin{aligned} 4 - \frac{11}{5} &\stackrel{?}{=} -3 + 4\left(\frac{8}{5}\right) \\ \frac{9}{5} &\neq \frac{17}{5} \quad \blacksquare \end{aligned}$$

Since we arrive at a contradiction, A and B do not intersect. Since they are also not parallel, A and B are *skew*.

Distances Between Skew Lines

Derivation: Distance between Two Skew Lines

1. Let lines L_1 and L_2 be skew and lie on planes P_1 and P_2 . P_1 must be parallel to P_2 . Assume L_1 is parallel to $\vec{v}_1$ and L_2 is parallel to $\vec{v}_2$. Finally, let M be a known point on L_1 and thus P_1 .
2. The normal vectors of both P_1 and P_2 must be the same as they are parallel. Since the normal vectors are orthogonal to both $\vec{v}_1$ and $\vec{v}_2$. We can find a normal vector by computing the cross product $\vec{n} = \vec{v}_1 \times \vec{v}_2$.
3. We can now completely define the plane P_1 using point M and normal vector $\vec{n}$. We can now find the distance from an arbitrary point N on L_2 to P_1 :

$$|L_1 L_2| = |\text{comp}_{\vec{v}_1 \times \vec{v}_2} \overrightarrow{MN}| = \frac{|(\vec{v}_1 \times \vec{v}_2) \cdot \overrightarrow{MN}|}{|\vec{v}_1 \times \vec{v}_2|}.$$

Example: Distance between Two Lines

Find the distance between the following two lines, assuming they are skew:

$$\begin{array}{lll} L_1 : & x(t) = 1 + t & y(t) = -2 + 3t & z(t) = 4 - t \\ L_2 : & x(s) = 2s & y(s) = 3 + s & z(s) = -3 + 4s \end{array}$$

The first step is to find the necessary parts needed to define a plane and a point.

Write out the direction vectors, $\vec{v}_1 = \langle 1, 3, -1 \rangle$ and $\vec{v}_2 = \langle 2, 1, 4 \rangle$. The normal vector to both planes containing L_1 and L_2 is:

$$\begin{aligned} \vec{n} &= \vec{v}_1 \times \vec{v}_2 = \langle 1, 3, -1 \rangle \times \langle 2, 1, 4 \rangle \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & -1 \\ 2 & 1 & 4 \end{vmatrix} = \langle 13, -6, -5 \rangle. \end{aligned}$$

We can now define the plane that contains L_1 using the normal vector $\vec{n}$ and a point that lies on L_1 , $\langle 1, -2, 4 \rangle$. Now, find a vector from a point on L_2 to a point on L_1 (and thus the plane containing it):

$$\vec{d} = \langle 0, 3, -3 \rangle - \langle 1, -2, 4 \rangle = \langle -1, 5, -7 \rangle$$

Finally, find the absolute value of the scalar projection of $\vec{d}$ along $\vec{n}$:

$$|L_1 L_2| = |\text{comp}_{\vec{n}} \vec{d}| = \frac{|\vec{n} \cdot \vec{d}|}{|\vec{n}|} = \frac{|\langle 13, -6, -5 \rangle \cdot \langle -1, 5, -7 \rangle|}{\sqrt{13^2 + (-6)^2 + (-5)^2}} = \frac{8}{\sqrt{230}}.$$

Extra: Distance from Point to Line

It can be shown that the distance from a point P to a line L with direction vector $\vec{v}$ is $\frac{|\vec{v} \times \overrightarrow{P_0 P}|}{|\vec{v}|}$.

1.6 Functions and Surfaces

- A scalar function f of n variables assigns each ordered collection of n real numbers $(x_1, x_2, \dots, x_n)$ in a domain $D \subseteq \mathbb{R}^n$ to a single real number $f(x_1, x_2, \dots, x_n)$.
The range of f can be written in set notation as $\{f(x_1, x_2, \dots, x_n) \mid (x_1, x_2, \dots, x_n) \in D\}$.
 - We say a scalar function f of n variables has a mapping $\mathbb{R}^n \mapsto \mathbb{R}$.
- Surfaces in 3-space are represented by functions of two variables, $z = f(x, y)$, where z is a dependent variable and x and y are independent variables.
- We can use **traces** to sketch a graph. By finding the curve of intersection between the function and a set of basic planes ($x = a$, $y = b$, and/or $z = c$), we can sketch the rough shape of a graph.
 - A **horizontal trace** is a trace along a plane $z = c$.
 - A **vertical trace** is a trace along one of the other basic planes.
- A **contour curve** or a **level curve** at k is the specific horizontal trace at $z = k$.
 - A **contour diagram**, where contour curves of varying values of k are overlaid on top of another, can be an alternative way to display a 3D surface in flat contexts.

3D Shapes

- A **cylinder** is defined to be any surface where the cross sectional curve along an axis is constant. In other words, it is a surface defined by at most two of the three variables.
The equation of a cylinder parallel to the xy -plane is $z = f(x)$ or $z = g(y)$.
 - An **elliptical cylinder** is a cylinder where the horizontal cross section is an ellipse.
 - A **parabolic cylinder** is a cylinder where the horizontal cross section is a parabola.
- A **quadric surface** is any surface defined by a second degree linear equation of x , y , or z .
 - In other words, it is a surface defined by an equation which only contains terms that are constant multiples of some second (or lower) degree combination of x , y , and z (ex. x^2 , y^2 , z^2 , xy , xz , yz).
General form of a quadric surface: $ax^2 + by^2 + cz^2 + dxy + exz + fyz + gx + hy + iz + j$ where $a \dots j$ are constants
 - An **ellipsoid** is a quadric surface where all traces are ellipses. A sphere is an example of an ellipsoid. They take the general form $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.
 - An **elliptic paraboloid** is a quadric surface where horizontal traces are ellipses, but vertical traces are parabolas. They take the form $\frac{z}{c} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$.
(Paraboloid opens on z -axis because z is the linear term)
 - A **hyperbolic paraboloid** is a paraboloid where horizontal traces are hyperbolas.
They take the form $\frac{z}{c} = \frac{x^2}{a^2} - \frac{y^2}{b^2}$.
 - A **hyperboloid** is a quadric surface where horizontal traces are ellipses and vertical traces are hyperbolas. Hyperboloids can have either one or two “sheets”.
They take the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = k$, where k can help determine the type of hyperboloid.
The hyperboloid will open along the axis whose term in the above equation is negative.
 - A hyperboloid has only one sheet if $k > 0$, and two sheets if $k < 0$.
 - A hyperboloid is a **cone** if $k = 0$. The sheets meet at a single point.

1.7 Alternative Coordinate Systems

Cylindrical Coordinates

In the **cylindrical coordinate system**, a point in $\mathbb{R}^3$ is represented by a polar coordinate as well as a z-coordinate, represented by (r, θ, z) . The conversion between cylindrical and rectangular coordinates is very similar to how it is done in 2D with polar coordinates:

$$\begin{aligned} (x, y, z) \rightarrow (r, \theta, z) : \quad r^2 &= x^2 + y^2 & \tan \theta &= \frac{y}{x} & z &= z \\ (r, \theta, z) \rightarrow (x, y, z) : \quad x &= r \cos \theta & y &= r \sin \theta & z &= z \end{aligned}$$

- Cylindrical coordinates are convenient to describe surfaces that are symmetrical about an axis.
- Cylindrical coordinates need not be oriented via z (e.g. we could have (r, θ, x))

Spherical Coordinates

In the **spherical coordinate system**, a point P in $\mathbb{R}^3$ is described using the radial distance from the point to the origin and two angles, represented by (ρ, θ, φ) , where:

- ρ is the **radial distance** to the origin, which can be computed as $\rho = |\vec{P}|$.
 - It is important to distinguish this from r , the **polar distance**, which is the projection of the radial distance to the xy-plane: $r = \rho \sin \theta$.
- θ is the **polar angle**, the angle used in polar coordinates, computed when P is projected onto the xy-plane.
- φ is the **azimuthal angle**, the angle between the +z-axis and the line connecting P to the origin.

It should also be noted that the restrictions $\rho \geq 0, 0 \leq \theta < 2\pi, 0 \leq \varphi < \pi$ can be used if we want a set of unique points.

Again, the line connecting P to the origin with magnitude ρ makes an angle φ with the +z-axis. We can derive that $r = \rho \sin \varphi$ and $z = \rho \cos \varphi$ are the components of this line. This gives us the conversion into cylindrical coordinates:

$$\begin{aligned} (r, \theta, z) \rightarrow (\rho, \theta, \varphi) : \quad \rho^2 &= r^2 + z^2 & \theta &= \theta & \tan \varphi &= \frac{r}{z} \\ (\rho, \theta, \varphi) \rightarrow (r, \theta, z) : \quad r &= \rho \sin \varphi & \theta &= \theta & z &= \rho \cos \varphi \end{aligned}$$

We can then substitute into the conversions between cylindrical and rectangular coordinates to get the conversions between spherical and rectangular coordinates:

$$\begin{aligned} (x, y, z) \rightarrow (\rho, \theta, \varphi) : \quad \rho^2 &= x^2 + y^2 + z^2 & \tan \theta &= \frac{y}{x} & \tan \varphi &= \frac{\sqrt{x^2 + y^2}}{z} \\ (\rho, \theta, \varphi) \rightarrow (x, y, z) : \quad x &= \rho \sin \varphi \cos \theta & y &= \rho \sin \varphi \sin \theta & z &= \rho \cos \varphi \end{aligned}$$

2 Vector Valued Functions

2.1 Vector Functions

- We've looked at functions which took n inputs and output one value, called scalar functions of n variables. We also said that these functions satisfy the mapping $\mathbb{R}^n \mapsto \mathbb{R}$.
- **Vector functions** (or vector-valued functions) are functions which take a certain amount of variables and output *many* values. These values are represented as vectors.

We say that a vector function $\vec{r}$ of m variables which outputs n values satisfies the mapping $\mathbb{R}^m \mapsto \mathbb{R}^n$. We will only be dealing with vector functions of one variable: $\vec{r}: \mathbb{R} \mapsto \mathbb{R}^n$.

- The components of a vector function $\vec{r}$ are known as the **component functions** of $\vec{r}$. For example, f , g , and h are the component functions of $\vec{r}$ if $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$.

Domain of a Vector Function

The domain D of a vector function $\vec{r}(t) = \langle f_1(t), f_2(t), \dots, f_n(t) \rangle$ is the **intersection** of the domains of $f_1, f_2, \dots, f_n$ in t :

$$D = \text{Dom}(\vec{r}(t)) = \bigcap_{k=1}^n \text{Dom}(f_k(t)).$$

Limit of a Vector Function

The limit of a vector function of one variable $\vec{r}(t) = \langle f_1(t), f_2(t), \dots, f_n(t) \rangle$ is the vector containing the limits of each component function of $\vec{r}$:

$$\lim_{t \rightarrow c} \vec{r}(t) = \left\langle \lim_{t \rightarrow c} f_1(t), \lim_{t \rightarrow c} f_2(t), \dots, \lim_{t \rightarrow c} f_n(t) \right\rangle.$$

Space Curves

Vector functions of one variable trace out **space curves** in 3-space. If all component functions of a vector function $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$ are continuous on some interval I , then we can define a **space curve** C as the set of points represented by the position vectors $\forall t \in I$.

The space curve C will have parametric equations: $x = f(t)$, $y = g(t)$, $z = h(t)$, and this can be shown by simply assuming $\vec{r}(t)$ represents the position vector of a point on C at any time t :

$$\vec{r}(t) = \langle x, y, z \rangle = \langle f(t), g(t), h(t) \rangle.$$

Intersections of Planes

- **The intersection of two surfaces is a space curve.** We know that two planes in 3-space can have a *line of intersection*. In a similar vein, any two surfaces in 3-space can have a **curve of intersection**, which is a space curve.

2.2 Calculus with Vector Functions

Derivatives of Vector Functions

The derivative $\vec{r}'(t)$ of a vector function $\vec{r}(t)$ is defined the same way as for normal functions:

$$\frac{d\vec{r}}{dt} = \vec{r}'(t) = \lim_{h \rightarrow 0} \frac{\vec{r}(t+h) - \vec{r}(t)}{h}.$$

Derivation: Derivative of a Vector Function by Components

Using basic laws of vectors and limits, we can also see that the derivative of a vector function is just a vector of the derivatives of each component function. If $\vec{r}(t) = \langle f_1(t), f_2(t), \dots, f_n(t) \rangle$, then:

$$\begin{aligned} \vec{r}'(t) &= \lim_{h \rightarrow 0} \frac{\langle f_1(t+h), f_2(t+h), \dots, f_n(t+h) \rangle - \langle f_1(t), f_2(t), \dots, f_n(t) \rangle}{h} \\ &= \lim_{h \rightarrow 0} \frac{\langle f_1(t+h) - f_1(t), f_2(t+h) - f_2(t), \dots, f_n(t+h) - f_n(t) \rangle}{h} \\ &= \lim_{h \rightarrow 0} \left\langle \frac{f_1(t+h) - f_1(t)}{h}, \frac{f_2(t+h) - f_2(t)}{h}, \dots, \frac{f_n(t+h) - f_n(t)}{h} \right\rangle \\ &= \left\langle \lim_{h \rightarrow 0} \frac{f_1(t+h) - f_1(t)}{h}, \lim_{h \rightarrow 0} \frac{f_2(t+h) - f_2(t)}{h}, \dots, \lim_{h \rightarrow 0} \frac{f_n(t+h) - f_n(t)}{h} \right\rangle \\ &= \langle f_1'(t), f_2'(t), \dots, f_n'(t) \rangle. \end{aligned}$$

Derivative of a Vector Function

Let the vector function $\vec{r}(t) = \langle f_1(t), f_2(t), \dots, f_n(t) \rangle$ have differentiable components. Then the derivative of $\vec{r}$, $\vec{r}'$ is a vector of the derivatives of each component:

$$\frac{d}{dt} \vec{r}(t) = \vec{r}'(t) = \langle f_1'(t), f_2'(t), \dots, f_n'(t) \rangle.$$

Similarly, the k th derivative of $\vec{r}$ is:

$$\frac{d^k}{dt^k} \vec{r}(t) = \vec{r}^{(k)}(t) = \langle f_1^{(k)}(t), f_2^{(k)}(t), \dots, f_n^{(k)}(t) \rangle$$

Remember that in 3-space, if the component functions of $\vec{r}(t)$ are continuous over an interval I , there exists a **space curve** C represented by $\vec{r}(t)$. For any given value of $t \in I$, $\vec{r}'(t)$ represents a vector tangent to C at the position represented by $\vec{r}(t)$. This vector is called the **tangent vector**.

Similarly, the **unit tangent vector** is often used to evaluate which direction the curve is moving in at any given $t \in I$, but not necessarily at what magnitude. This vector is represented:

$$\text{Unit Tangent Vector} = \vec{T}(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}.$$

Differentiation Rules for Vector Functions

If $\vec{u}(t)$ and $\vec{v}(t)$ are differentiable vector functions, then:

- Linearity Rule: $\frac{d}{dt}(a\vec{u}(t) + b\vec{v}(t)) = a\vec{u}'(t) + b\vec{v}'(t)$ where a and b are constants.
- Scalar Product Rule: $\frac{d}{dt}f(t)\vec{u}(t) = f'(t)\vec{u}(t) + f(t)\vec{u}'(t)$ where $f(t)$ is a scalar function.
- Dot Product Rule: $\frac{d}{dt}(\vec{u}(t) \cdot \vec{v}(t)) = \vec{u}'(t) \cdot \vec{v}(t) + \vec{u}(t) \cdot \vec{v}'(t)$.
- Cross Product Rule: $\frac{d}{dt}(\vec{u}(t) \times \vec{v}(t)) = \vec{u}'(t) \times \vec{v}(t) + \vec{u}(t) \times \vec{v}'(t)$.
- Chain Rule: $\frac{d}{dt}\vec{u}(f(t)) = \vec{u}'(f(t))f'(t)$ where $f(t)$ is a scalar function.

Integrals of Vector Functions

Definite Integral of a Vector Function

The definite integral from $t = a$ to $t = b$ of a vector function $\vec{r}(t) = \langle f_1(t), f_2(t), \dots, f_n(t) \rangle$ is the vector consisting of the definite integrals from $t = a$ to $t = b$ over each of its components:

$$\int_a^b \vec{r}(t) dt = \left\langle \int_a^b f_1(t) dt, \int_a^b f_2(t) dt, \dots, \int_a^b f_n(t) dt \right\rangle.$$

Antiderivative of a Vector Function

The antiderivative $\vec{R}(t)$ of a vector function $\vec{r}(t) = \langle f_1(t), f_2(t), \dots, f_n(t) \rangle$ is one where $\vec{R}'(t) = \vec{r}(t)$. It is the vector consisting of the antiderivatives of component functions, but it can vary by a constant vector $\vec{C}$ (since each component can vary independently):

$$\vec{R}(t) = \int \vec{r}(t) dt = \left\langle \int f_1(t) dt, \int f_2(t) dt, \dots, \int f_n(t) dt \right\rangle + \vec{C}.$$

2.3 Arc Length and Curvature

Arc Length

Derivation: Arc Length of a Space Curve

Let's derive the arc length s of a space curve C in $\mathbb{R}^3$ defined by $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$ from $t = a$ to $t = b$.

We know that an infinitesimally small movement along the curve C , $d\vec{r}$, is represented by their respective movements in the x , y , and z -axes:

$$d\vec{r} = \langle dx, dy, dz \rangle.$$

The magnitude of each small movement $d\vec{r}$ results in a small contribution to arc length:

$$\begin{aligned} ds = |d\vec{r}| &= \sqrt{dx^2 + dy^2 + dz^2} \\ &= \sqrt{\left(\frac{dx}{dt} dt\right)^2 + \left(\frac{dy}{dt} dt\right)^2 + \left(\frac{dz}{dt} dt\right)^2} \\ &= \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt \end{aligned}$$

We can finally add up all small changes in arc length from $t = a$ to $t = b$ using an integral:

$$s = \int_a^b ds = \int_a^b |\vec{r}'(t)| dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt.$$

Arc Length of a Space Curve

Let the space curve C be defined by the vector function $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$. Then the **arc length** s of C from $t = a$ to $t = b$ is:

$$\text{Arc Length} = s = \int_a^b |\vec{r}'(t)| dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt.$$

Parameterizations

- We know that a space curve C can be defined by a vector-valued function represented by a set of parametric equations, but there exists an infinite set of parametric equations for C called the **parameterizations** of C .

For example, let the curve C be defined using the vector function $\vec{r}(t) = \langle t, t^2, \sin t \rangle$. We can apply the transformation $t = e^u$ to get a new curve D , $\vec{r}(u) = \langle e^u, e^{2u}, \sin(e^u) \rangle$. Since there exists a bijection from t to e^u , all points on C are also points on D , and vice-versa, so $C \equiv D$.

- We can often make the definition of a curve independent of parameterization by defining it in terms of “how far we are along the curve”, i.e. arc length:

Arc Length Function of a Curve

If the curve C is defined using the vector function $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$, its **arc length function** $\vec{R}(s)$ is parameterization of $\vec{r}$ using s , where $s(t)$ is the arc length from some starting point $t = a$ to t :

$$s(t) = \int_a^t |\vec{r}'(\tau)| d\tau = \int_a^t \sqrt{\left(\frac{dx}{d\tau}\right)^2 + \left(\frac{dy}{d\tau}\right)^2 + \left(\frac{dz}{d\tau}\right)^2} d\tau.$$

It holds that $\frac{ds}{dt} = |\vec{r}'(t)|$ by the fundamental theorem of calculus. We can use this differential equation to determine a transformation from $t \rightarrow s$.

Example: Reparameterization using Arc Length

The curve C is defined by $\vec{r}(t) = \langle 3t, 5 - 4t, 6 + 2t \rangle$.

Write an *arc length function* $\vec{q}(s)$ for C .

First, we know that $\vec{r}'(t) = \langle 3, -4, 2 \rangle$ and thus $|\vec{r}'(t)| = \sqrt{29}$.

We can find the arc length s as a function of t , choosing $s = 0$ at $t = 0$:

$$\begin{aligned} s(t) &= \int_0^t |\vec{r}'(\tau)| d\tau = \int_0^t \sqrt{29} d\tau = \sqrt{29}t \\ &\Rightarrow t = \frac{s}{\sqrt{29}}. \end{aligned}$$

After solving for t in terms of s , we perform the substitution, and the parameterization is:

$$\vec{q}(s) = \left\langle \frac{3}{\sqrt{29}}s, 5 - \frac{4}{\sqrt{29}}s, 6 + \frac{2}{\sqrt{29}}s \right\rangle.$$

Curvature

Curvature is a measure of how quickly a space curve changes direction at any given point. The measure is not concerned about the magnitude of change; only direction.

Recall that the direction of the tangent vector can be represented using the **unit tangent vector**, $\vec{T}(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}$. Then the curvature κ is how quickly this vector changes with respect to where we are on the curve, $\kappa = \left| \frac{d\vec{T}}{ds} \right|$.

Different parameterizations of t can produce the same curve (for example $\langle t, 2t, 3t \rangle$ and $\langle 2t, 4t, 6t \rangle$), but the progression along the curve may be slower or faster if t increases at given rate. Thus, we use the *arc length* as a measure of how far we are on the curve.

Curvature of a Space Curve

Let the space curve C be defined by the vector function $\vec{r}(t)$. If an arc length parameterization $\vec{T}(s)$ of the unit tangent vector exists, then the **curvature** κ of C is:

$$\text{Curvature} = \kappa = \left| \frac{d\vec{T}}{ds} \right|.$$

If $\vec{T}(t)$ is the unit vector tangent to $\vec{r}(t)$ at t , then the curvature κ of C at t is:

$$\kappa(t) = \frac{|\vec{T}'(t)|}{|\vec{r}'(t)|} = \frac{|\vec{r}'(t) \times \vec{r}''(t)|}{|\vec{r}'(t)|^3}.$$

Special Case: Curvature of 2D Curve

If C completely lies on a plane and can be defined by $y = f(x)$, then we can parameterize this into a vector valued function $\vec{r}(x) = \langle x, f(x), 0 \rangle$.

Then $\vec{r}'(x) = \langle 1, f'(x), 0 \rangle$, $\vec{r}''(x) = \langle 0, f''(x), 0 \rangle$, and $\vec{r}'(x) \times \vec{r}''(x) = \langle 0, 0, f''(x) \rangle$:

$$\kappa(x) = \frac{|f''(x)|}{(1 + (f'(x))^2)^{3/2}}.$$

Example: Computing the Curvature of a Space Curve

Find the curvature of the space curve defined by $\vec{r}(t) = t\hat{i} + t^2\hat{j} + t^3\hat{k}$ as a function of t .

Since we are only given $\vec{r}$, it would be most convenient to use the formula in terms of only $\vec{r}$ and its derivatives.

Compute the derivatives of $\vec{r}$:

$$\vec{r}(t) = \langle t, t^2, t^3 \rangle$$

$$\vec{r}'(t) = \langle 1, 2t, 3t^2 \rangle$$

$$\vec{r}''(t) = \langle 0, 2, 6t \rangle.$$

Compute $|\vec{r}' \times \vec{r}''|$:

$$\begin{aligned} |\vec{r}' \times \vec{r}''| &= \left\| \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2t & 3t^2 \\ 0 & 2 & 6t \end{vmatrix} \right\| \\ &= |\langle 6t^2, -6t, 2 \rangle| \\ &= \sqrt{36t^4 + 36t^2 + 4} \\ &= 2\sqrt{9t^4 + 9t^2 + 1}. \end{aligned}$$

Compute $|\vec{r}'|^3$:

$$\begin{aligned} |\vec{r}'|^3 &= |\langle 1, 2t, 3t^2 \rangle|^3 \\ &= \left| \sqrt{1 + 4t^2 + 9t^4} \right|^3 \\ &= (1 + 4t^2 + 9t^4)^{3/2}. \end{aligned}$$

The quotient is the curvature:

$$\kappa = \frac{|\vec{r}' \times \vec{r}''|}{|\vec{r}'|^3} = \frac{2\sqrt{9t^4 + 9t^2 + 1}}{(1 + 4t^2 + 9t^4)^{3/2}}.$$

Derivation: Curvature as functions of t

By the chain rule:

$$\kappa = \left| \frac{d\vec{T}}{ds} \right| = \left| \frac{d\vec{T}/dt}{ds/dt} \right| = \frac{|\vec{T}'(t)|}{|\vec{r}'(t)|}.$$

By the quotient rule:

$$\vec{T}'(t) = \frac{\vec{r}''(t)|\vec{r}'(t)| - \vec{r}'(t)\frac{d}{dt}|\vec{r}'(t)|}{|\vec{r}'(t)|^2}.$$

Recall that $\frac{d|x|}{dx} = \frac{x}{|x|}$. The same can be applied to vector magnitudes:

$$\begin{aligned} \vec{T}'(t) &= \frac{\vec{r}''(t)|\vec{r}'(t)| - \vec{r}'(t)\frac{d}{dt}|\vec{r}'(t)|}{|\vec{r}'(t)|^2} \\ &= \frac{\vec{r}''(t)|\vec{r}'(t)|^2 - \vec{r}'(t)(\vec{r}'(t) \cdot \vec{r}''(t))}{|\vec{r}'(t)|^3} \end{aligned}$$

Recall the a property of vector triple products, $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$.

Since $|\vec{r}'(t)|^2 = \vec{r}'(t) \cdot \vec{r}'(t)$:

$$\begin{aligned} \vec{r}''(t)|\vec{r}'(t)|^2 - \vec{r}'(t)(\vec{r}'(t) \cdot \vec{r}''(t)) &= (\vec{r}'(t) \cdot \vec{r}'(t))\vec{r}''(t) - (\vec{r}'(t) \cdot \vec{r}''(t))\vec{r}'(t) \\ &= \vec{r}'(t) \times (\vec{r}''(t) \times \vec{r}'(t)). \end{aligned}$$

Since $\vec{r}''(t) \times \vec{r}'(t)$ is orthogonal to $\vec{r}'(t)$, the following holds:

$$\begin{aligned} |\vec{r}'(t) \times (\vec{r}''(t) \times \vec{r}'(t))| &= |\vec{r}'(t)| |\vec{r}''(t) \times \vec{r}'(t)| \\ &= |\vec{r}'(t)| |\vec{r}'(t) \times \vec{r}''(t)|. \end{aligned}$$

Finally,

$$\begin{aligned} |\vec{T}'(t)| &= \frac{|\vec{r}'(t)| |\vec{r}'(t) \times \vec{r}''(t)|}{|\vec{r}'(t)|^3} = \frac{|\vec{r}'(t) \times \vec{r}''(t)|}{|\vec{r}'(t)|^2}. \\ \kappa(t) &= \frac{|\vec{T}'(t)|}{|\vec{r}'(t)|} = \frac{|\vec{r}'(t) \times \vec{r}''(t)|}{|\vec{r}'(t)|^2} \frac{1}{|\vec{r}'(t)|} = \frac{|\vec{r}'(t) \times \vec{r}''(t)|}{|\vec{r}'(t)|^3}. \end{aligned}$$

Normal Vectors and Planes

- We know that $\vec{T}$, the unit tangent vector, will always have a magnitude of 1. Thus, $|\vec{T}|$ is always equal to 1. Since $|\vec{T}|$ is constant, then $|\vec{T}'| = 0$ and $\vec{T} \cdot \vec{T}' = |\vec{T}| |\vec{T}'| \cos \theta = 0$. Thus, $\vec{T}$ is always orthogonal to its derivative $\vec{T}'$.

The unit vector of $\vec{T}'$ is known as the **principal unit normal vector**, $\vec{N}(t) = \frac{\vec{T}'(t)}{|\vec{T}'(t)|}$.

Proof: A vector function $\vec{r}$ is orthogonal to $\vec{r}'$ if $\vec{r}$ has a constant length

If the statement above can be proved true, then we can be certain that because the unit tangent vector $\vec{T}$ has a constant length 1, it must be orthogonal to $\vec{T}'$, and since $\vec{N}$ is parallel to $\vec{T}'$, $\vec{N}$ is orthogonal to $\vec{T}$ and thus $\vec{r}$.

Let $\vec{r}(t)$ be a vector function in n -space with components $\vec{r}(t) = \langle f_1(t), f_2(t), \dots, f_n(t) \rangle$. Then the tangent vector function is $\vec{r}'(t) = \langle f_1'(t), f_2'(t), \dots, f_n'(t) \rangle$.

If $\vec{r}(t)$ has a constant magnitude, then $f_1^2(t) + f_2^2(t) + \dots + f_n^2(t) = c$, where c is a constant.

By implicit differentiation:

$$\begin{aligned} f_1^2(t) + f_2^2(t) + \dots + f_n^2(t) &= c \\ 2f_1(t)f_1'(t) + 2f_2(t)f_2'(t) + \dots + 2f_n(t)f_n'(t) &= 0 \\ f_1(t)f_1'(t) + f_2(t)f_2'(t) + \dots + f_n(t)f_n'(t) &= 0 \\ \vec{r}(t) \cdot \vec{r}'(t) &= 0. \quad \blacksquare \end{aligned}$$

This is exactly the dot product of $\vec{r}(t)$ and $\vec{r}'(t)$. The dot product of two orthogonal vectors is 0, so $\vec{r}$ must be orthogonal to $\vec{r}'$ at all times if $\vec{r}$ has a constant magnitude.

- The **binormal vector** $\vec{B}$ is the unit vector orthogonal to both $\vec{T}$ and $\vec{N}$ at the same time, so it can be represented by $\vec{B}(t) = \vec{T}(t) \times \vec{N}(t)$. Since both $\vec{T}$ and $\vec{N}$ are unit vectors, $\vec{B}$ is also a unit vector.

The TNB Frame

A **normal unit vector** $\vec{n}$ to a space curve $\vec{r}(t)$ at t is orthogonal to $\vec{T}(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}$, the unit tangent vector at t . We typically deal with two normal unit vectors: the **principal normal unit vector** or **unit normal** $\vec{N}(t)$ and the **binormal vector** $\vec{B}(t)$, where:

$$\vec{N}(t) = \frac{\vec{T}'(t)}{|\vec{T}'(t)|} \quad \vec{B}(t) = \vec{T}(t) \times \vec{N}(t).$$

The **TNB frame** is the set of unit vectors $\vec{T}$, $\vec{N}$, and $\vec{B}$ which define how a curve is behaving at a point, which are all components of the derivatives of $\vec{r}$, where:

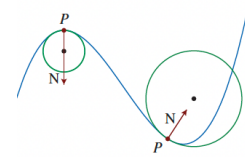
- $\vec{T}$ represents how the particle is *moving*, quantified using *speed* $v = \left| \frac{d\vec{r}}{dt} \right|$
- $\vec{N}$ represents how the particle is *turning*, quantified using *curvature*, $\kappa = \left| \frac{d\vec{T}}{ds} \right|$
- $\vec{B}$ represents how the particle is *twisting*, quantified using *torsion*, $\tau = -\frac{d\vec{B}}{ds} \cdot \vec{N}$.

- The plane P which contains both $\vec{N}$ and $\vec{B}$ is called the **normal plane** because it is normal to the tangent vector $\vec{T}$, meaning the vector normal to P is parallel to $\vec{T}$.
- The **rectifying plane** R is the plane which contains both $\vec{T}$ and $\vec{B}$, lying normal to $\vec{N}$.
- The **osculating plane** O is the plane that contains both $\vec{T}$ and $\vec{N}$, and is the plane which lies close to the curve near the point of tangency. The osculating plane is normal to $\vec{B}$.
- If we know the tangent and normal vectors to a space curve C at a point P , we can determine the **circle of curvature** or **osculating circle** at P , which approximates the behavior C using a

circle (as opposed to a tangent line). They can be useful as circles of curvature consider the curvature κ in the approximation:

Circle of Curvature

Let the space curve C have curvature κ at a point P . Then the **circle of curvature** or **osculating circle** at P is the circle which (1) lies on the osculating plane of C , (2) contains the point P , (3) has radius $R = \frac{1}{\kappa}$, and (4) has a center Q a distance $\frac{1}{\kappa}$ from P along the principal normal vector $\vec{N}$:



$$\vec{Q} = \vec{P} + \frac{\vec{N}}{\kappa} \quad R = \frac{1}{\kappa}$$

R is known as the **radius of curvature**.

Extension: Torsion

Torsion is a quantity τ which measures how fast the osculating plane rotates as we move along the curve (i.e. with respect to arc length s). Since the osculating plane is described using its normal vector $\vec{B}$, we can suppose that τ should be expressed in terms of $\frac{d\vec{B}}{ds}$.

Torsion of a Space Curve

Let a space curve $\vec{r}$ have the TNB unit vectors $\vec{T}, \vec{N}, \vec{B}$ at some point P on the curve. Then the **torsion** τ at P is:

$$\text{Torsion} = \tau = -\frac{d\vec{B}}{ds} \cdot \vec{N}.$$

With Respect to Time

The torsion can also be found using the following formula:

$$\tau(t) = \frac{\det(\vec{r}, \vec{r}', \vec{r}'')}{|\vec{r}' \times \vec{r}''|^2}.$$

Example: Torsion of a Circular Helix

Find the torsion of the circular helix $\vec{r}(t) = \langle a \cos t, a \sin t, bt \rangle$ as a function of t , where a and b are constants.

First, find relevant derivatives:

$$\vec{r}' = \langle -a \sin t, a \cos t, b \rangle$$

$$\vec{r}'' = \langle -a \cos t, -a \sin t, 0 \rangle$$

$$\vec{r}''' = \langle a \sin t, -a \cos t, 0 \rangle.$$

Then, $\det(\vec{r}', \vec{r}'', \vec{r}''')$ is the same as:

$$\begin{vmatrix} -a \sin t & a \cos t & b \\ -a \cos t & -a \sin t & 0 \\ a \sin t & -a \cos t & 0 \end{vmatrix} = a^2 b.$$

Find: $|\vec{r}' \times \vec{r}''|^2$:

$$\begin{aligned} |\vec{r}' \times \vec{r}''|^2 &= \left\| \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -a \sin t & a \cos t & b \\ -a \cos t & -a \sin t & 0 \end{vmatrix} \right\|^2 = |\langle ab \sin t, ab \cos t, a^2 \sin^2 t + a^2 \cos^2 t \rangle|^2 \\ &= |\langle ab \sin t, ab \cos t, a^2 \rangle|^2 \\ &= a^2 b^2 \sin^2 t + a^2 b^2 \cos^2 t + a^4 = a^2(b^2 + 1). \end{aligned}$$

Take the quotient to find torsion:

$$\tau = \frac{a^2 b}{a^2(b^2 + 1)} = \frac{b}{b^2 + 1}.$$

Summary

Arc Length: $s = \int_a^b |\vec{r}'(t)| dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$

Curvature (3D): $\kappa = \left| \frac{d\vec{T}}{ds} \right| = \frac{|\vec{T}'(t)|}{|\vec{r}'(t)|} = \frac{|\vec{r}' \times \vec{r}''|}{|\vec{r}'|^3}$

Curvature (2D): $\kappa = \frac{|f''(x)|}{(1 + (f'(x))^2)^{3/2}}$

Radius of Curvature: $r = \frac{1}{\kappa}$

TNB Frame: $\vec{T} = \frac{\vec{r}'}{|\vec{r}'|} \quad \vec{N} = \frac{\vec{T}'}{|\vec{T}'|} \quad \vec{B} = \vec{T} \times \vec{N}$

$\vec{T}$ normal to normal plane; $\vec{N}$ normal to rectifying plane; $\vec{B}$ normal to osculating plane

Torsion: $\tau = -\frac{d\vec{B}}{ds} \cdot \vec{N} = \frac{\det(\vec{r}', \vec{r}'', \vec{r}''')}{|\vec{r}' \times \vec{r}''|}$

2.4 Applications of Vector Functions

Particle Motion Definitions

If $\vec{r}(t)$ is the **position vector** of a particle at any time t , then:

- $\vec{v}(t) = \vec{r}'(t)$ is the **velocity vector** at time t
 - Its magnitude $v(t) = |\vec{v}(t)| = |\vec{r}'(t)|$ is the **speed** at time t
 - We can also take the derivative of *distance traveled*, s , over time: $v(t) = s'(t) = \frac{d}{dt}|\vec{r}(t)|$.
From this, we can see that $|\frac{d}{dt}\vec{r}(t)| = \frac{d}{dt}|\vec{r}(t)|$. This is a special property of the position vector.
- $\vec{a}(t) = \vec{v}'(t) = \vec{r}''(t)$ is the **acceleration vector** at time t .

Tangential and Centripetal Acceleration

- We have been able to come up with a scalar analog of velocity, *speed*, which is the derivative of arc length or the magnitude of velocity.
- However, the same cannot be said for acceleration. The derivative of speed is not necessarily equal to the magnitude of acceleration. In other words, $|\vec{v}'(t)| \neq v'(t)$.
- When we only consider speed, the changing direction of the velocity vector is lost. The function $\vec{v}(t)$ would be the same whether we were moving straight at a constant rate or whether we are moving around in a circle at a constant rate.
- Thus, we must consider acceleration using at least two scalar components in order to properly represent the potential to change direction. When evaluating particle dynamics, it is useful to break down acceleration into two useful components:
 - **tangential acceleration**, a_t , which is the component of acceleration directed tangent to the curve (in the same direction as $\vec{T}$)
 - **centripetal acceleration**, a_c , which is the component of acceleration responsible for changing the direction of the particle, directed towards the center of the circle of curvature, which will always be normal to the curve (in the same direction as $\vec{N}$).

Components of Acceleration

The acceleration vector $\vec{a}$ can be broken down into two components.

- **Tangential Acceleration** a_t is the derivative of speed, $a_t = \frac{dv}{dt} = \frac{d^2s}{dt^2} = \frac{\vec{r}' \cdot \vec{r}''}{|\vec{r}'|}$.
- **Centripetal Acceleration** a_c is $a_c = \kappa v^2 = \frac{|\vec{r}' \times \vec{r}''|}{|\vec{r}'|^3}$, where κ is curvature.

If $\vec{T}$ is the unit tangent vector to the curve at t and $\vec{N}$ is the unit normal at t , then:

$$\vec{a} = a_t \vec{T} + a_c \vec{N} = v' \vec{T} + \kappa v^2 \vec{N}.$$

Tangential and Centripetal Acceleration

Since the tangent unit vector $\vec{T}(t)$ is just the unit vector of $\vec{v}(t)$, representing only the direction of the velocity, we can also write $\vec{v}(t) = v(t)\vec{T}(t)$ (*velocity = speed in the direction of $\vec{T}$*)

From this, we can write a new expression for $\vec{a}(t)$. Since $\vec{a}(t) = \vec{v}'(t)$, then:

$$\vec{a} = \frac{d}{dt}(v\vec{T}) = v'\vec{T} + v\vec{T}'.$$

We have already established in 2.3 that $\vec{T}'$ will always be orthogonal to $\vec{T}$, which is why its unit vector is known as the **normal vector** $\vec{N} = \frac{\vec{T}'}{|\vec{T}'|}$. So, we see that we have broken down $\vec{a}$ into a component pointing tangent to the curve and a component pointing normal to the curve.

Since we are considering vector *components*, a_t and a_c are scalar values. We can now represent $\vec{a}(t)$ using a_t and a_c pointing in their respective directions:

$$\vec{a} = a_t\vec{T} + a_c\vec{N}.$$

We can also rewrite the first equation by using the fact that:

- $\kappa = \frac{|\vec{T}'|}{|\vec{r}'|} \Rightarrow |\vec{T}'| = \kappa|\vec{r}'| = \kappa v$, where κ is the curvature at t
- $\vec{N} = \frac{\vec{T}'}{|\vec{T}'|} \Rightarrow \vec{T}' = |\vec{T}'|\vec{N} = \kappa v\vec{N}$

$$\vec{a} = v'\vec{T} + \kappa v^2\vec{N}$$

Thus, we find that the components of acceleration are:

$$a_t = v' \qquad a_c = \kappa v^2 = \frac{|\vec{r}' \times \vec{r}''|}{|\vec{r}'|^3}.$$

Since curvature is the reciprocal of the radius of curvature R , we can also write:

$$a_c = \frac{v^2}{R}.$$

This is the equation typically introduced in a mechanics course.

2.5 Parametric Surfaces

- A surface can be parameterized, but it must be done so using two variables, as points on the surface can vary independently in two “dimensions”. Similarly, we would be able to parameterize a volume using three variables, and so on.
- Two *parameters*, say, u and v , can independently vary throughout their domains, with each combination producing a point $P(x, y, z)$ on a surface S . Each coordinate of P is a function of both parameters: $P(x(u, v), y(u, v), z(u, v))$. Then the parametric equations of S are:

$$x = x(u, v) \qquad y = y(u, v) \qquad z = z(u, v).$$

- We can also represent a parametric surface using a vector function of two variables. For S as defined above, we can also use $\vec{r}(u, v) = x(u, v)\hat{i} + y(u, v)\hat{j} + z(u, v)\hat{k}$. As u and v vary, the function $\vec{r}$ outputs the position vector for some point on S .

Grid Curves

A method of graphing a parametric surface is by finding the **grid curves** of the surface. Each curve is achieved by setting one of the parameters to a constant value. This results in a parametric equation of only one parameter, which are space curves.

For example, from the surface $\vec{r}(u, v) = \langle u^2 + v^2, u, v \rangle$, we can deduce the following grid curves:

- When $u = 0$, $\vec{r}(v) = \langle v^2, 0, v \rangle$ is the parabola $x = z^2$, opening along the $+x$ -axis
- When $u = k$, $\vec{r}(v) = \langle k^2 + v^2, k, v \rangle$ is the parabola above shifted k^2 units to the right and k units along the $+y$ -axis
- When $v = 0$, $\vec{r}(u) = \langle u^2, u, 0 \rangle$ is a parabola opening to the right on the plane $z = 0$
- When $v = k$, $\vec{r}(u) = \langle u^2 + k^2, u, k \rangle$ is a parabola opening to the right, shifted k^2 units right and k units up, along the $+z$ -axis

Parametric Equations of a Plane

To find the parametric equation of a plane, find a point $\vec{r}_0$ which lies on the plane as well as two nonparallel vectors $\vec{u}$ and $\vec{v}$ which lie on the plane. We can step an arbitrary amount μ and ν in each of the direction of the vectors to reach every point on the plane.

Since the vectors $\vec{u}$ and $\vec{v}$ are linearly independent, they can be considered basis vectors for the plane.

Vector and Parametric Equations of a Plane

Let the plane P contain point $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$ and the vectors $\vec{u} = \langle u_1, u_2, u_3 \rangle$ and $\vec{v} = \langle v_1, v_2, v_3 \rangle$ lie on P , and $\vec{u}$ is not parallel to $\vec{v}$.

Then a vector equation of P is

$$\vec{r}(\mu, \nu) = \vec{r}_0 + \mu\vec{u} + \nu\vec{v}.$$

This can be decomposed into the following parametric equations:

$$x(\mu, \nu) = x_0 + u_1\mu + v_1\nu$$

$$y(\mu, \nu) = y_0 + u_2\mu + v_2\nu$$

$$z(\mu, \nu) = z_0 + u_3\mu + v_3\nu.$$

3 Multivariate Limits and Derivatives

3.1 Graphs of Multivariate Functions

- The **level curves** of a surface $z = f(x, y)$ are the curves of intersection at the level planes $z = k$, or in other words, the curves defined by $f(x, y) = k$. A level curve may not exist for all values of k .
- Similarly, the **level surfaces** of a volume $w = f(x, y, z)$ are the surfaces defined by $f(x, y, z) = k$.
- The **contour plot** is a way of representing a 3D surface on a 2D plane, by drawing all level curves on one plane. In other words, the contour plot of a surface $z = f(x, y)$ can be drawn by drawing lines which join the points (x, y) where $f(x, y)$ is equal.

3.2 Limits and Continuity in Multiple Variables

Summary

- **To show a limit does not exist**, check multiple paths and show result is $\neq$
- **To show and find a limit exists:**
 - Direct substitution (if function is continuous over its domain)
 - Algebraic manipulation (cancel factors/multiply by conjugates)
 - Squeeze Theorem
 - Convert to polar/spherical coordinates, then check if limit is constant
 $(x, y) \rightarrow (0, 0)$ becomes $r \rightarrow 0^+$, $(x, y, z) \rightarrow (0, 0, 0)$ becomes $\rho \rightarrow 0^+$
 - Use $\delta - \varepsilon$ proof

Limits & Continuity

- Recall that for single-variable functions: $\lim_{x \rightarrow a} f(x) = L$ means that by approaching $x = a$ from both the left and right, $f(x)$ approaches the same value L . When $\lim_{x \rightarrow a} f(x) = f(a)$, the function is *continuous*.
- For a function of two variables $f(x, y)$, we can approach some point $(x, y) = (a, b)$ from an infinite number of paths. If f approaches the same limit L , then $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$.
 - To show a limit exists, we must prove that by approaching the point from all paths, we get the same limit. If even one of the paths yield a different limit than another path, the limit does not exist.
 - f is **continuous** at (a, b) if $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b)$.

We can use this definition in reverse to find limits of continuous functions (i.e. if we know a function is continuous and defined at (a, b) , then we can just plug in (a, b) into the function to find its limit)

Example: Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + \sin^2 y}{2x^2 + y^2}$ does not exist.

Check that approaching the limit from at least two paths yield different values:

From the line $y = 0$:

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ y \rightarrow 0}} \frac{x^2 + \sin^2 y}{2x^2 + y^2} = \lim_{x \rightarrow 0} \frac{x^2}{2x^2} = \frac{1}{2}.$$

From the line $x = 0$:

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ x \rightarrow 0}} \frac{x^2 + \sin^2 y}{2x^2 + y^2} = \lim_{y \rightarrow 0} \frac{\sin^2 y}{y^2} = \left(\lim_{y \rightarrow 0} \frac{\sin y}{y} \right)^2 = 1.$$

$$\therefore \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + \sin^2 y}{2x^2 + y^2} \text{ DNE.}$$

Squeeze Theorem

- Let there be three functions of two variables α, f, β .

If, for a finite region D , $\alpha(x, y) \leq f(x, y) \leq \beta(x, y)$, then for any point (a, b) in D where

$\lim_{(x,y) \rightarrow (a,b)} \alpha(x, y) = \lim_{(x,y) \rightarrow (a,b)} \beta(x, y) = L$, the **squeeze theorem** says that $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$.

Extension: Limits by Changing Coordinate Systems

- If our multivariate limit approaches the origin, we can convert our function into polar coordinates (for two variables) or spherical coordinates (for three variables).

This decays our limit into a single-variable limit (the limit as distance from the origin approaches zero).

- For polar coordinates: as $(x, y) \rightarrow (0, 0)$, $r \rightarrow 0^+$.
For spherical coordinates: as $(x, y, z) \rightarrow (0, 0, 0)$, $\rho \rightarrow 0^+$.

Rectangular to Polar Coordinates

Let $g(r, \theta)$ be the polar transformation of $f(x, y)$, i.e. $g(r, \theta) = f(r \cos \theta, r \sin \theta)$. Then:

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{r \rightarrow 0^+} g(r, \theta).$$

Rectangular to Spherical Coordinates

Let $g(\rho, \theta, \varphi)$ be the spherical transformation of $f(x, y, z)$, i.e.

$g(\rho, \theta, \varphi) = f(\rho \sin \varphi \cos \theta, \rho \sin \varphi \sin \theta, \rho \cos \varphi)$. Then:

$$\lim_{(x,y,z) \rightarrow (0,0,0)} f(x, y, z) = \lim_{\rho \rightarrow 0^+} g(\rho, \theta, \varphi).$$

- θ (or φ) is the angle of the path we approach the origin from. If our limit depends on θ or φ , our limit would depend on the angle of the path and thus our limit does not exist.

Example: Limit using Polar Coordinates

Determine if $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$ exists.

Converting into polar coordinates:

$$\frac{x^2 - y^2}{x^2 + y^2} \mapsto \frac{r^2(\cos^2 \theta - \sin^2 \theta)}{r^2} = \cos 2\theta.$$

As $(x, y) \rightarrow (0, 0)$, $r \rightarrow 0^+$. Thus, our limit is equivalent to:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2} = \lim_{r \rightarrow 0^+} \cos 2\theta = \cos 2\theta.$$

The limit depends on θ , which is the angle of the path we approach the origin from. In other words, the limit is different depending on the path, meaning our limit **does not exist**.

Extension: Delta-Epsilon Definitions of Limits

The definition of a single-variable limit is:

$$\lim_{x \rightarrow a} f(x) = L \text{ means } \forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

In other words, for any desired vertical distance ε , we must be able to find an x a horizontal distance δ away from $x = a$ where evaluating $f(x)$ is at most ε away from $y = L$.

We can extend this to 3D: if ε is the desired "error", then there must exist a point (x, y) a radial distance δ away from (a, b) where evaluating $f(x, y)$ is at most ε away from L .

Delta-Epsilon Definition of a Multivariate Limit

Let $f(x, y)$ be a function of two variables. If $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$, then:

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } 0 < \sqrt{(x - a)^2 + (y - b)^2} < \delta \implies |f(x, y) - L| < \varepsilon.$$

3.3 Partial Derivatives

- Recall that the derivative $f'(x)$ of a single-variable function $f(x)$ at some point $x = a$ is the rate of change of the output f as the input x changes near $x = a$.
- For a function of multiple independent variables, the variables can change independently. One variable can change at once, or multiple can change at the same time.
- The **partial derivative** $\frac{\partial f}{\partial x} = f_x(x, y, \dots)$ of a function of several variables $f(x, y, \dots)$ is the rate of change of f as only one of the variables (in this case x) changes, and the other variables are held constant.
 - If f is a function of x and y , the partial derivative w.r.t. x of f at a point $(x, y) = (a, b)$, $\left. \frac{\partial f}{\partial x} \right|_{(a,b)} = f_x(a, b)$ is the rate of change of f as x varies while other y is held constant near (a, b) .
- Notations:** $\frac{\partial f}{\partial x} = f_x = \partial_x f$ (Leibniz, Lagrange, and Euler notations respectively)

Limit Definition of a Partial Derivative

For Functions of Two Variables

The **partial derivative** of a function $f(x, y)$ with respect to x is:

$$\frac{\partial f}{\partial x} = f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}.$$

Similarly, the partial derivative of f with respect to y is:

$$\frac{\partial f}{\partial y} = f_y(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h}.$$

It is the change in f as the variable we are differentiating with respect to changes while all other variables are held constant.

General Case

Let $f(x_1, x_2, \dots, x_n)$ be a function of n variables. Then the **partial derivative** of f with respect to x_i is:

$$\frac{\partial f}{\partial x_i} = f_{x_i}(x_1, x_2, \dots, x_n) = \lim_{h \rightarrow 0} \frac{f(x_1, x_2, \dots, x_i + h, \dots, x_n) - f(x_1, x_2, \dots, x_n)}{h}.$$

Example: Find the Partial Derivatives of $f(x, y, z) = xe^y + y^2e^z + ze^{-x}$.

To find each partial derivative, differentiate normally, treating other variables constant.

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(xe^y + y^2e^z + ze^{-x}) = e^y + -ze^{-x}.$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(xe^y + y^2e^z + ze^{-x}) = xe^y + 2ye^z.$$

$$\frac{\partial f}{\partial z} = \frac{\partial}{\partial z}(xe^y + y^2e^z + ze^{-x}) = y^2e^z + e^{-x}.$$

Higher-Order Partial Derivatives

A **second-order partial derivative** of a function of two variables $f(x, y)$ is the partial derivative of one of the partial derivatives of f . This means that there are **four** total second-order partial derivatives of f :

$$\begin{aligned} f_{xx} &= \frac{\partial^2 f}{\partial x^2} & f_{xy} &= \frac{\partial^2 f}{\partial y \partial x} \\ f_{yx} &= \frac{\partial^2 f}{\partial x \partial y} & f_{yy} &= \frac{\partial^2 f}{\partial y^2} \end{aligned}$$

Generally, a function of n variables has n^2 second-order partial derivatives.

Note

For Leibniz notation, the order of differentiation is read right to left, whereas with the subscript (Lagrange) notation, the order is from left to right.

Clairaut's Theorem

Suppose a function $f(x, y)$ is *defined* over a finite region D . If f_{xy} and f_{yx} are *continuous* throughout D , then for all points $(x, y) \in D$:

$$f_{xy} = f_{yx} \quad \text{or} \quad \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}.$$

In other words, for continuous functions, the order of partial derivatives does not matter. This theorem can be extended to higher-order derivatives.

This property is often referred to as the **symmetry of second partial derivatives**.

- In a similar vein, a **third-order partial derivative** of a function is a partial derivative of one of the second-order partial derivatives of f . Generally, a function of n variables has n^3 third-order partial derivatives.
e.g. $f(x, y)$ has 8 third-order partial derivatives: $f_{xxx}, f_{xxy}, f_{xyx}, f_{xyy}, f_{yxx}, f_{yxy}, f_{yyx}, f_{yyy}$.
- An n th-order partial derivative of a function of m variables is a partial derivative of one of the $(n - 1)$ th-order partial derivatives of f . Generally, a function of m variables has m^n n th-order partial derivatives.

Differentiability

Differentiability by Partial Derivatives

A function $f(x, y)$ is **differentiable** at a point (a, b) if f is both **continuous** on some closed finite region D which includes (a, b) and the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are both **defined** on the open region D .

This can be extended to more than 2 variables.

Partial Differential Equations

- An equation consisting of partial derivatives is called a **partial differential equation** (PDE). For example, $\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}$ is a partial differential equation called the *wave equation*.
- A **solution** to a partial differential equation is a function which satisfies the equation.
 - To verify a solution, find necessary partial derivatives and plug them into the equation to see if it holds true.

Example: Verifying a solution to a PDE

Verify that the function $u(x, y) = e^x \sin y$ is a solution to the PDE $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

Find the necessary partial derivatives:

$$\begin{aligned}\frac{\partial u}{\partial x} &= e^x \sin y & \frac{\partial u}{\partial y} &= e^x \cos y \\ \frac{\partial^2 u}{\partial x^2} &= e^x \sin y & \frac{\partial^2 u}{\partial y^2} &= -e^x \sin y.\end{aligned}$$

Plugging into the PDE:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = e^x \sin y - e^x \sin y \stackrel{\checkmark}{=} 0.$$

3.4 Tangent Planes & Approximations

Recall that the derivative of a single-variable function can be interpreted as the *slope of the tangent line* to the curve. A function of two variables defines a surface, there are an infinite number of tangent lines to the surface at any given point, however there exists a *plane* that consists of all tangent lines. This plane is called the **tangent plane**.

Consider a curve defined by single-variable function $y = f(x)$. Recall that the linearization of f at a point $x = x_0$ can be used to approximate f at points near x_0 . Such linearization is:

$$y - y_0 = f'(x)(x - x_0).$$

Tangent Plane to a Surface

Let the surface S be defined by the equation $z = f(x, y)$. Then the **tangent plane** to S at a point $(x_0, y_0, f(x_0, y_0))$ is the plane defined by:

$$z - f(x_0, y_0) = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

We can use this tangent plane to derive a general function called the *linearization function* which can be used to approximate values of f near the point:

Linearization of a Function of Two Variables

The **linearization** of a function $f(x, y)$ at (x_0, y_0) is:

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

The **linear approximation** of a function $f(x, y)$ at a point P is the linearization at that point.

Example: Linear Approximation

Use the linearization of $f(x, y) = \sqrt{x^2 + y^2}$ at $(1, 1)$ to approximate $f(1.1, 1.2)$.

Find the partial derivatives of f :

$$\frac{\partial f}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}} \quad \frac{\partial f}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}}$$

Plugging into each at $(1, 1)$ yields $f_x(1, 1) = f_y(1, 1) = \frac{\sqrt{2}}{2}$.

Find the linearization using the formula above:

$$f(x, y) \approx L(x, y) = \sqrt{2} + \frac{\sqrt{2}}{2}(x - 1) + \frac{\sqrt{2}}{2}(y - 1).$$

Thus:

$$f(1.1, 1.2) \approx L(1.1, 1.2) = \sqrt{2} + \frac{\sqrt{2}}{2}(0.1) + \frac{\sqrt{2}}{2}(0.2) = \frac{23\sqrt{2}}{20}.$$

Differentials

Recall the tangent plane equation. If the point of interest (x_0, y_0, z_0) is made infinitesimally close to an arbitrary point (x, y, z) , then the difference of each component is also infinitesimally small. That is, if $x - x_0 \rightarrow 0$, we can call this difference dx , called the **differential** of x .

Thus, our equation becomes:

$$dz = f_x dx + f_y dy = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy.$$

So if $z = f(x, y)$, the above expression dz is the **total differential** of z .

Total Differential

Let f be a function of n variables: $f(x_1, x_2, \dots, x_n)$. Then the **total differential** of f is:

$$df = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \dots + \frac{\partial f}{\partial x_n} dx_n.$$

Example: Find the total differential of $V(r, h) = \frac{1}{3}\pi r^2 h$.

The partial derivatives of V are:

$$\frac{\partial V}{\partial r} = \frac{2}{3}\pi r h \qquad \frac{\partial V}{\partial h} = \frac{1}{3}\pi r^2$$

Thus, the total differential is:

$$dV = \frac{2}{3}\pi r h dr + \frac{1}{3}\pi r^2 dh.$$

Using a finite interpretation of a differential, we interpret let each differential as an “error” or finite difference. Then, the total differential becomes an equation relating relatively small changes in each of the variables to the total change in function.

Example: Interpreting differentials as finite errors

A ruler is used to measure the height of a cone to an accuracy of 3 mm. A separate ruler is used to measure the radius of the cone to an accuracy of 4 mm.

The cone is measured this way to have a height of 14 mm and a radius of 6 mm, resulting in an estimated volume of $168\pi \text{ mm}^3$. What is the approximate max error of the estimate?

Using the total differential from the other example, we can estimate the error dV from the measurement. Since dr can be interpreted as the “error in r ”, we can set $dr = 4 \text{ mm}$. Similarly, $dh = 3 \text{ mm}$. Finally, $r = 6 \text{ mm}$ and $h = 14 \text{ mm}$.

Plugging all four values into the expression for dV :

$$\Delta V_{\max} \approx dV = \frac{2}{3}\pi(6 \text{ mm})(14 \text{ mm})(4 \text{ mm}) + \frac{1}{3}\pi(6 \text{ mm})^2(3 \text{ mm}) = 260\pi \text{ mm}^3.$$

Differentials vs. Increments

The **increment** of a function f , represented Δf is the *true* change in f over some change in its variables $\vec{r}$, represented $\Delta \vec{r}$. A **differential** uses a tangent line to approximate the increment ($df \approx \Delta f$), and the approximation gets better as $d\vec{r} \rightarrow \vec{0}$.

Tangent Planes to Parametric Surfaces

- Recall that a parametric surface is defined by a vector function of two variables, for example $\vec{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle$.
- The tangent plane to a parametric surface at a point P can be found using the cross product of the two tangent vectors $\frac{\partial \vec{r}}{\partial u}$ and $\frac{\partial \vec{r}}{\partial v}$.
 - Originating from P , all tangent vectors can be thought of as “lying on the tangent plane”
 - Thus normal vector $\vec{n}$ to the tangent plane is orthogonal to any two tangent vectors at P . We can choose the two most convenient ones, $\frac{\partial \vec{r}}{\partial u}$ and $\frac{\partial \vec{r}}{\partial v}$:

$$\vec{n} = \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v}.$$

- If $\vec{n} = \langle a, b, c \rangle$, then the equation of the tangent plane at point $P(x_0, y_0, z_0)$ is $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$ by the standard normal-point form of a plane.

Example: Tangent Plane to a Parametric Surface

Find the equation of the plane tangent to the surface defined by $\vec{r}(u, v) = \langle u \cos v, u \sin v, v \rangle$ at the point $(0, 1, \frac{\pi}{2})$.

First, find the values of u and v at the point:

$$u \cos v = 0 \qquad u \sin v = 1 \qquad v = \frac{\pi}{2}.$$

From the first two equations, we can see that $u = 1$ and $v = \frac{\pi}{2}$.

Next, find the tangent vectors:

$$\frac{\partial \vec{r}}{\partial u} = \vec{r}_u = \langle \cos v, \sin v, 0 \rangle \qquad \frac{\partial \vec{r}}{\partial v} = \vec{r}_v = \langle -u \sin v, u \cos v, 1 \rangle.$$

Plugging in $u = 1$ and $v = \frac{\pi}{2}$ yields:

$$\vec{r}_u \left(1, \frac{\pi}{2} \right) = \langle 0, 1, 0 \rangle \qquad \vec{r}_v \left(1, \frac{\pi}{2} \right) = \langle -1, 0, 1 \rangle.$$

The normal vector is:

$$\vec{n} = \vec{r}_u \times \vec{r}_v = \langle 0, 1, 0 \rangle \times \langle -1, 0, 1 \rangle = \langle 1, 0, 1 \rangle.$$

Finally, plug the point and normal vector into the point-normal form of a plane:

$$1(x - 0) + 0(y - 1) + 1\left(z - \frac{\pi}{2}\right) = 0 \quad \text{or} \quad x + z = \frac{\pi}{2}.$$

Extension: Quadratic Approximations

Recall: the quadratic approximation formula for a single-variable function $f(x)$ from a known point $(a, f(a))$ is just the second-degree Taylor polynomial centered at $x = a$:

$$f(x) \approx Q(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2.$$

We can extend this to functions of two variables:

$$\begin{aligned} f(x, y) \approx Q(x, y) = & f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b) \\ & + \frac{f_{xx}(a, b)}{2}(x - a)^2 + f_{xy}(a, b)(x - a)(y - b) + \frac{f_{yy}(a, b)}{2}(y - b)^2. \end{aligned}$$

- The first term is just the function plugged in at the point, just like the single-variable approximation.
- The next two terms are just the linearization of f at (a, b) . These are the parts of the approximation involving the “first-derivative”, just like the single-variable approximation.
- The last three terms are the parts of the approximation involving the “second-derivative”. There are actually four terms, but since $f_{xy} = f_{yx}$, the two mixed partial derivative terms can be combined (hence why there is no division by 2)

3.5 Chain Rule

The **chain rule** computes the derivative of a function of other functions.

- For single-variable functions $y = f(x)$, $\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$ if there is an “intermediate function” $u(x)$ where y is a function of u .

- For multivariate functions $f(x, y, \dots)$, there are two “cases” of composite functions:

- **Case 1:** if each variable is a function of a single variable, e.g. $x = g(t)$, $y = h(t)$, then:

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \dots$$

- **Case 2:** if each variable is a function of several variables, e.g. $x = g(u, v)$, $y = h(u, v)$, then f has partial derivatives in both u and v :

$$\frac{\partial f}{\partial u} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u} + \dots$$

$$\frac{\partial f}{\partial v} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial v} + \dots$$

Example: Chain Rule

Let $z = x^2 + y^2 + xy$, where $x = \ln t$ and $y = \cos t$. What is $\frac{dz}{dt}$?

By the formula for case 1:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}.$$

The partial derivatives of z are:

$$\frac{\partial z}{\partial x} = 2x + y \qquad \frac{\partial z}{\partial y} = 2y + x.$$

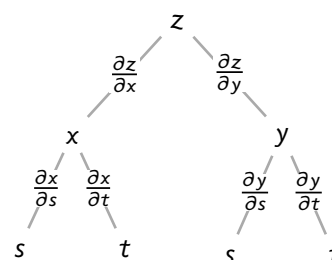
The derivatives of x and y are:

$$\frac{dx}{dt} = \frac{1}{t} \qquad \frac{dy}{dt} = -\sin t.$$

Thus:

$$\frac{dz}{dt} = (2x + y) \left(\frac{1}{t} \right) + (2y + x)(-\sin t) = \frac{2 \ln t + \cos t}{t} - (2 \cos t + \ln t) \sin t.$$

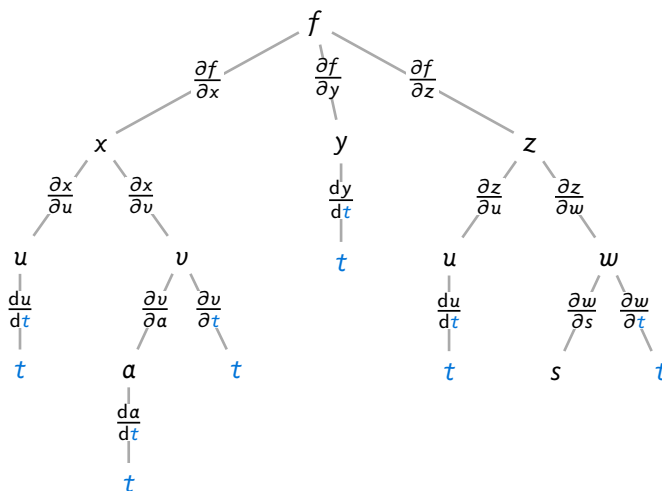
- We can use a **tree diagram** to help visualize the chain rule. Each node is a variable, and each edge represents a partial derivative.
 - The path from the root to a leaf node represents the full “chain” between variables.
 - Find all paths that connect the root to the leaf node of interest, and sum the products of each path.



Example: Chain Rule for Tree Diagram

Use a tree diagram to find a formula for $\frac{\partial f}{\partial t}$ if the following functions are given:
 $f(x, y, z)$, $x(u, v)$, $y(t)$, $z(u, w)$, $u(t)$, $v(a, t)$, $a(t)$, and $w(s, t)$.

Find all nodes with t , then trace each path leading to each node:



There are 6 paths from f to t :

1. $f \rightarrow x \rightarrow u \rightarrow t$ becomes the "chain" $\frac{\partial f}{\partial x} \frac{\partial x}{\partial u} \frac{du}{dt}$.
"f is a function of x, which is a function of u, which is a function of t"
2. $f \rightarrow x \rightarrow v \rightarrow a \rightarrow t$
3. $f \rightarrow x \rightarrow v \rightarrow t$
4. $f \rightarrow y \rightarrow t$
5. $f \rightarrow z \rightarrow u \rightarrow t$
6. $f \rightarrow z \rightarrow w \rightarrow t$

Thus, by the chain rule:

$$\frac{\partial f}{\partial t} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} \frac{du}{dt} + \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} \frac{\partial v}{\partial a} \frac{da}{dt} + \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} \frac{\partial v}{\partial t} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial u} \frac{du}{dt} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial w} \frac{\partial w}{\partial t}.$$

Implicit Differentiation

- A function can be either **explicitly** defined (e.g. $z = f(x, y)$) or **implicitly** defined (e.g. $F(x, y, z) = 0$, where z is a function of x and y).
 - For example, $\sin(y) = x^2$ implicitly defines y as a function of x .
- To differentiate an implicitly defined function, we use the chain rule just like with single-variable functions.
 - Recall: if $y(x)$ is implicitly defined by $\sin(y) = x^2$, we can find $\frac{dy}{dx}$ by:
 1. taking $\frac{d}{dx}$ of both sides: $\frac{d}{dx} \sin(y) = 2x$
 2. using the chain rule if we see a y : $\frac{d}{dx} \sin(y) = \frac{d(\sin(y))}{dy} \frac{dy}{dx} = \cos(y) \frac{dy}{dx}$
 3. solving for $\frac{dy}{dx}$: $\frac{dy}{dx} = \frac{2x}{\cos(y)}$.
 - If $z(x, y)$ is implicitly defined, then there are two derivatives we can find: $\frac{\partial z}{\partial x}$ or $\frac{\partial z}{\partial y}$.

- To find $\frac{\partial z}{\partial x}$, take $\frac{\partial}{\partial x}$ of both sides, use the chain rule when you see a z , then solve for $\frac{\partial z}{\partial x}$.
- To find $\frac{\partial z}{\partial y}$, take $\frac{\partial}{\partial y}$ of both sides, use the chain rule when you see a z , then solve for $\frac{\partial z}{\partial y}$.

Example: Implicit Differentiation

Let $z(x, y)$ be a function of x and y . If $x^3 + y^3 + z^3 + 6xyz = 1$, what is $\frac{\partial z}{\partial x}$?

To find $\frac{\partial z}{\partial x}$, take the partial derivative w.r.t. x of both sides:

$$\begin{aligned}\frac{\partial}{\partial x}(x^3 + y^3 + z^3 + 6xyz) &= \frac{\partial}{\partial x}(1) \\ 3x^2 + 0 + \frac{\partial}{\partial x}(z^3) + \frac{\partial}{\partial x}(6xyz) &= 0.\end{aligned}$$

Let $u = z^3$. To find $\frac{\partial}{\partial x}(z^3) = \frac{\partial u}{\partial x}$, we must use the chain rule, since u is not an independent variable. Since u is a function of z is a function of x , the chain rule tells us that:

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial z} \frac{\partial z}{\partial x}.$$

Since $\frac{\partial u}{\partial z} = 3z^2$, $\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}(z^3) = 3z^2 \frac{\partial z}{\partial x}$. Similarly we can find using the product rule that $\frac{\partial}{\partial x}(6xyz) = 6y \frac{\partial}{\partial x}(xz) = 6yz + 6xy \frac{\partial z}{\partial x}$:

$$\begin{aligned}3x^2 + 3z^2 \frac{\partial z}{\partial x} + 6yz + 6xy \frac{\partial z}{\partial x} &= 0 \\ \frac{\partial z}{\partial x} &= -\frac{3x^2 + 6yz}{3z^2 + 6xy} = -\frac{x^2 + 2yz}{z^2 + 2xy}.\end{aligned}$$

Implicit Function Theorem

We can also use partial differentiation directly to find the derivative of implicitly-defined functions (alternative to implicit differentiation).

Let $y(x)$ be implicitly defined by the equation $F(x, y) = 0$. By taking the partial derivative of both sides w.r.t. x , we get:

$$\begin{aligned}\frac{\partial}{\partial x}(F(x, y)) &= 0 \\ \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} \frac{dy}{dx} &= 0 \\ \frac{dy}{dx} &= -\frac{\partial F / \partial x}{\partial F / \partial y}.\end{aligned}$$

Implicit Function Theorem for Single-Variable Functions

Let $y = f(x)$ be defined implicitly by the equation $F(x, y) = 0$. Then:

$$\frac{dy}{dx} = -\frac{\partial F / \partial x}{\partial F / \partial y}.$$

Example: If $x^2 + y^2 + xy = 7$, what is $\frac{dy}{dx}$?

Let $F(x, y) = x^2 + y^2 + xy - 7$. The partial derivatives of F are:

$$\frac{\partial F}{\partial x} = 2x + y \qquad \frac{\partial F}{\partial y} = 2y + x.$$

Thus:

$$\frac{dy}{dx} = -\frac{\partial F/\partial x}{\partial F/\partial y} = -\frac{2x + y}{2y + x}.$$

If we have a function $z(x, y)$ implicitly defined by the equation $F(x, y, z) = 0$, we can find both $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ by taking the partial derivative of both sides w.r.t. x and y , respectively:

$$\begin{aligned} \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial x} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial x} &= 0 \implies \frac{\partial z}{\partial x} = -\frac{\partial F/\partial x}{\partial F/\partial z}. \\ \frac{\partial F}{\partial x} \frac{\partial x}{\partial y} + \frac{\partial F}{\partial y} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial y} &= 0 \implies \frac{\partial z}{\partial y} = -\frac{\partial F/\partial y}{\partial F/\partial z}. \end{aligned}$$

Since x is independent of y , $\frac{\partial y}{\partial x} = 0$ and $\frac{\partial x}{\partial y} = 0$.

Implicit Function Theorem for Multivariable Functions

Let $z = f(x, y)$ be defined implicitly by the equation $F(x, y, z) = 0$. Then:

$$\frac{\partial z}{\partial x} = -\frac{\partial F/\partial x}{\partial F/\partial z} \qquad \frac{\partial z}{\partial y} = -\frac{\partial F/\partial y}{\partial F/\partial z}.$$

Example: Implicit Function Theorem for Multivariable Functions

Let $z(x, y)$ be defined implicitly by the equation $x^3 + y^3 + z^3 + 6xyz = 1$. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

Let $F(x, y, z) = x^3 + y^3 + z^3 + 6xyz - 1$. The partial derivatives of F are:

$$\frac{\partial F}{\partial x} = 3x^2 + 6yz \qquad \frac{\partial F}{\partial y} = 3y^2 + 6xz \qquad \frac{\partial F}{\partial z} = 3z^2 + 6xy.$$

Thus:

$$\frac{\partial z}{\partial x} = -\frac{\partial F/\partial x}{\partial F/\partial z} = -\frac{3x^2 + 6yz}{3z^2 + 6xy} \qquad \frac{\partial z}{\partial y} = -\frac{\partial F/\partial y}{\partial F/\partial z} = -\frac{3y^2 + 6xz}{3z^2 + 6xy}.$$

This is the same answer we got using implicit differentiation.

3.6 Gradients

The **gradient** is a vector of all partial derivatives of a function. If f is a function that takes in n parameters, then the gradient of f , typically denoted $\vec{\nabla}f$, is an n -dimensional vector.

Gradient of a Function

Let f be a function of n variables: $f(x_1, x_2, \dots, x_n)$. Then the **gradient** of f is:

$$\vec{\nabla}f(x_1, x_2, \dots, x_n) = \left\langle \frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right\rangle.$$

For example, $\vec{\nabla}f(x, y) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$ and $\vec{\nabla}f(x, y, z) = \langle f_x, f_y, f_z \rangle$.

The *nabla* operator, $\vec{\nabla}$, can be interpreted as a vector of partial derivative operators:

$$\vec{\nabla} = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle.$$

This is why $\vec{\nabla}f$ can be used to represent the gradient of f :

$$\vec{\nabla}f = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle.$$

This definition of $\vec{\nabla}$ will be helpful in future units.

Total Differential in terms of the Gradient

Take the total differential of a function $f(x, y)$, $df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$. We can represent this as the following dot product:

$$df = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle \cdot \langle dx, dy \rangle.$$

If we let $d\vec{r} = \langle dx, dy \rangle$ (a change in “position”), then the total differential can be written as just:

$$df = \vec{\nabla}f \cdot d\vec{r}.$$

Directional Derivatives

- Suppose a surface $z = f(x, y)$ is graphed in the 3D-plane. Each partial derivative is the slope of the tangent line in the direction of the respective axis. For example, $\frac{\partial z}{\partial x}$ is the slope of the tangent line in the direction of $+x$.
- With partial derivatives, we can only represent the slope in two directions ($+x$ and $+y$). To represent the slope in any direction, we use a **directional derivative**.
- The directional derivative $D_{\vec{v}}f(x, y)$ is the slope of the line tangent to the surface *in the direction of $\vec{v}$* is denoted $D_{\vec{v}}f(x, y)$.

Directional Derivative

Let $z = f(x, y)$ be a surface in the 3D-plane, and $\hat{v} = \langle v_1, v_2 \rangle$ is a 2D unit vector. The **directional derivative** in the direction of $\hat{v}$ is:

$$D_{\hat{v}} f(x, y) = \vec{\nabla} f(x, y) \cdot \hat{v} = \frac{\partial f}{\partial x} v_1 + \frac{\partial f}{\partial y} v_2.$$

- Notice that the definition above only works when $\vec{v}$ is a unit vector.
- An n -dimensional manifold has a directional derivative in the direction of an $(n - 1)$ -dimensional vector (e.g. a 3D shape has a 2D dimensional gradient)
- The notation $\nabla_{\vec{v}}$ is sometimes used to denote $D_{\vec{v}}$. (e.g. $\nabla_{\vec{v}} f$).

Properties of Gradients

Direction of Steepest Ascent

- The direction for which the directional derivative is highest (direction of **steepest ascent**) is in the direction of the gradient vector. The maximum value of the directional derivative is the magnitude of the gradient:

$$\max D_{\hat{v}} f(x, y) = |\vec{\nabla} f(x, y)|.$$

- In the same way, the direction of **steepest descent** is in the direction of $-\vec{\nabla} f(x, y)$, and the minimum value of the directional derivative is $-|\vec{\nabla} f(x, y)|$.
- If the angle between the direction vector and the gradient vector is θ , then by the definition of the dot product,

$$D_{\hat{v}} f(x, y) = |\vec{\nabla} f(x, y)| |\hat{v}| \cos \theta = |\vec{\nabla} f(x, y)| \cos \theta.$$

- Note: this θ is **not** the angle between the x-axis and the direction vector!

Normal Lines and Tangent Planes

- The gradient vector at a point is **orthogonal** (perpendicular) to the level curve of the function at that point (or level surface if f is a function of 3 variables).
 - A level curve of f is in the form $f(x, y) = c$, where c is a constant. A level surface of f is in the form $f(x, y, z) = c$.
- The **tangent line** to a level *curve* using $\vec{\nabla} f$ is $\vec{\nabla} f(\vec{r}_0) \cdot (\vec{r} - \vec{r}_0) = 0$.
- We can find the **normal line** and **tangent plane** to a level *surface* using $\vec{\nabla} f$:
 - The normal line at point $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$ is given by:

$$\vec{r} = \vec{r}_0 + t \vec{\nabla} f(\vec{r}_0).$$

- The tangent plane at $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$ is given by:

$$\vec{\nabla} f(\vec{r}_0) \cdot (\vec{r} - \vec{r}_0) = 0.$$

Generalization of a Linear Approximation

Using the gradient vector, we can generalize the linear approximation for a multivariate function f , given its “center” $\vec{r}_0 = \langle x_0, y_0, z_0, \dots \rangle$ (i.e. the “given point”):

$$L(\vec{r}) = f(\vec{r}_0) + \vec{\nabla} f(\vec{r}_0) \cdot (\vec{r} - \vec{r}_0).$$

This is a first-order Taylor polynomial for f . It is similar to the formula for single-variable functions, but instead of derivatives, gradients are used.

Note that we pass in a position vector $\vec{r} = \langle x, y, z, \dots \rangle$ to represent all parameters passed into f .

Extension: The Hessian Matrix

If represent $\vec{\nabla}$ as a column vector and perform matrix multiplication:

Hessian Operator (Gradient Definition)

The **hessian operator** $\mathbf{H}$, when applied to a function, returns a matrix of its second-order partial derivatives. The operator is defined as:

$$\mathbf{H} = \vec{\nabla} \vec{\nabla}^T.$$

For example, the Hessian operator applied to a function $f(x, y)$ is:

$$\mathbf{H}(f) = \vec{\nabla} \vec{\nabla}^T f = \begin{pmatrix} \partial_x \\ \partial_y \end{pmatrix} (\partial_x \ \partial_y) f = \begin{pmatrix} f_{xx} & f_{yx} \\ f_{xy} & f_{yy} \end{pmatrix}.$$

The resulting matrix is known as the **Hessian matrix** of f :

Hessian Matrix

Let $f(x_1, x_2, \dots, x_n)$ be a function of n variables. Then the **Hessian matrix** of f is the matrix of all partial derivatives:

$$\mathbf{H}(f) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}$$

Or, more compactly, it is the $n \times n$ matrix $\mathbf{H}$ where $\mathbf{H}_{i,j} = \frac{\partial^2 f}{\partial x_i \partial x_j}$.

The Hessian matrix can be thought of as a “second-order gradient” and the analog to the second derivative in the Taylor polynomial.

Thus, the quadratic approximation of f at $\vec{r}_0$ is:

$$Q(\vec{r}) = f(\vec{r}_0) + \vec{\nabla} f(\vec{r}_0) \cdot (\vec{r} - \vec{r}_0) + \frac{1}{2} (\vec{r} - \vec{r}_0)^T \mathbf{H}(f(\vec{r}_0)) (\vec{r} - \vec{r}_0).$$

3.7 Extreme Values

Local Extrema

- A function $f(x, y)$ has a *local maximum* at (a, b) if $f(a, b) \geq f(x, y)$ for all points (x, y) near (a, b) .
- A function $f(x, y)$ has a *local minimum* at (a, b) if $f(a, b) \leq f(x, y)$ for all points (x, y) near (a, b) .
- Collectively, these are called *local extrema*.

Just like univariate functions, multivariate functions can only attain local extrema at **critical points**. This is where the function has a derivative that is either zero or undefined:

Critical Points

Let f be a function of several variables: $f(x, y, \dots)$. f has a **critical point** at $(a, b, \dots)$ if $\vec{\nabla}f(a, b, \dots) = \vec{0}$ or it is not differentiable at $(a, b, \dots)$ (i.e. $\vec{\nabla}f$ is undefined).

A local extrema does not exist at every critical point. To determine if a critical point is a local extrema, we can use the *second partial derivative test*:

Second Partial Derivative Test

Let $f(x, y)$ be a function of two variables whose second-order partial derivatives are **continuous**. Let $D = \det(\mathbf{H}(f))$ be the determinant of the *Hessian matrix* of f .

In other words: $D = \begin{vmatrix} f_{xx} & f_{yx} \\ f_{xy} & f_{yy} \end{vmatrix} = f_{xx}f_{yy} - f_{xy}^2$.

If (a, b) is a **critical point** of f , then:

- if $D(a, b) > 0$:
 - ...and $f_{xx} > 0$, then there is a *local minimum* at (a, b) .
 - ...and $f_{xx} < 0$, then there is a *local maximum* at (a, b) .
- if $D(a, b) < 0$, there is a *saddle point* at (a, b) .
- if $D(a, b) = 0$, the test is *inconclusive*.

Example: Finding Local Extrema using Second Partial Derivative Test

Find and classify all local extrema of the function $f(x, y) = x^3 + y^3 - 3x - 12y + 5$.

First, find the critical points by setting $\vec{\nabla}f = \vec{0}$:

$$\vec{\nabla}f(x, y) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle = \langle 3x^2 - 3, 3y^2 - 12 \rangle.$$

Setting each component equal to zero gives:

$$\begin{array}{ll} 3x^2 - 3 = 0 & 3y^2 - 12 = 0 \\ x = \pm 1 & y = \pm 2. \end{array}$$

Thus, the critical points are $(1, 2), (1, -2), (-1, 2), (-1, -2)$.

Next, compute the second-order partial derivatives:

$$f_{xx} = 6x \qquad f_{yy} = 6y \qquad f_{xy} = 0.$$

Thus, $D = |\mathbf{H}(f)| = \begin{vmatrix} 6x & 0 \\ 0 & 6y \end{vmatrix} = 36xy$.

Now evaluate D at each critical point:

(x, y)	$D(x, y)$	$f_{xx}(x, y)$	Classification
$(1, 2)$	36	6	Local Minimum since $D > 0$ and $f_{xx} > 0$
$(1, -2)$	-36		Saddle Point since $D < 0$
$(-1, 2)$	-36		Saddle Point since $D < 0$
$(-1, -2)$	36	-6	Local Maximum since $D > 0$ and $f_{xx} < 0$

Absolute Extrema

- A function $f(x, y)$ as an *absolute maximum* at (a, b) on a region D if $f(a, b) \geq f(x, y)$ for all points (x, y) in D .
- A function $f(x, y)$ as an *absolute minimum* at (a, b) on a region D if $f(a, b) \leq f(x, y)$ for all points (x, y) in D .
- Collectively, these are called *absolute extrema*.

Extreme Value Theorem

Let $f(x, y)$ be a function of two variables that is **continuous** on a *closed* and *bounded* region D . Then f attains both an absolute maximum and an absolute minimum on D .

Candidates Test

Recall the candidates test for single-variable functions: to find absolute extrema on a closed interval $[a, b]$:

1. Evaluate the function at all critical points in (a, b)
2. Evaluate the function at the endpoints $x = a$ and $x = b$.
3. The largest value is the absolute maximum, and the smallest value is the absolute minimum.

For multivariate functions $f(x, y)$ on a closed and bounded region D :

1. Evaluate the function at all critical points in the interior of D (where $\vec{\nabla}f = \vec{0}$ or undefined).
2. Evaluate the function on all candidates on the *boundary* of D .
3. The largest value is the absolute maximum, and the smallest value is the absolute minimum.

Essentially, the absolute extrema can occur either in the interior of the region (which we can easily find candidates for using $\vec{\nabla}f = \vec{0}$) or on the boundary of the region. Unlike single-variable functions, there are an infinite number of points at the boundary.

We must find a way to find candidates on the boundary which *may* contain the extrema.

For step two, there are two primary methods of finding these candidates:

- Find the absolute extrema of f on parts of the boundary where we can reduce f to a single-variable function (e.g. along line segments) — this is called the *closed interval method*.
- Use the method of **Lagrange Multipliers** to find candidates on more complex boundaries (this is the next section)

Example: Closed Interval Method with Line Segments

Find the absolute extrema of $f(x, y) = x^2 - 2xy + 2y$ on the triangular region with vertices at $(0, 0)$, $(2, 0)$, $(0, 2)$.

First, find the critical points in the interior of the region by setting $\vec{\nabla}f = \vec{0}$:

$$\vec{\nabla}f(x, y) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle = \langle 2x - 2y, -2x + 2 \rangle.$$

This is a linear system with only one solution: $(1, 1)$. This is our first *candidate*.

Next, evaluate f on each side of the triangle (L_1 , L_2 , and L_3):

- L_1 is the line segment from $(0, 0)$ to $(2, 0)$. Here, $y = 0$, so $f(x, 0) = x^2$.

To find the extrema on L_1 we consider critical points *and* endpoints as candidates.

- $f(x, 0)$ has a critical point at $x = 0$ since it satisfies $\frac{d}{dx}f(x, 0) = 2x = 0$. We must also consider endpoints at $x = 0$ and $x = 2$. Add $(0, 0)$, $(2, 0)$ as candidates.

- L_2 is the line segment from $(2, 0)$ to $(0, 2)$. Here, $y = -x + 2$, so $f(x, -x + 2) = x^2 - 2x(-x + 2) + 2(-x + 2) = 3x^2 - 6x + 4$.

There is a critical point at $x = 1$ ($\frac{d}{dx}f(x, -x + 2) = 6x - 6 = 0 \implies x = 1$) and endpoints at $x = 0$ and $x = 2$. Add $(1, 1)$, $(2, 0)$, $(0, 2)$ as candidates.

- L_3 is the line segment from $(0, 2)$ to $(0, 0)$. Here, $x = 0$, so $f(0, y) = 2y$.

There is no critical point and endpoints at $y = 0$ and $y = 2$. Add $(0, 0)$, $(0, 2)$ as candidates.

Now evaluate f at all candidates:

Source	(x, y)	$f(x, y)$
$\vec{\nabla}f = \vec{0}$ and L_2	$(1, 1)$	1
L_1, L_3	$(0, 0)$	0
L_1, L_2	$(2, 0)$	4
L_2, L_3	$(0, 2)$	4

The absolute minimum is 0 at $(0, 0)$ and the absolute maximum is 4 at $(2, 0)$ and $(0, 2)$.

3.8 Lagrange Multipliers

- When trying to find absolute extrema in a closed region, we had to find the minima and maxima of a known *boundary*.
 - For boundaries of single variable functions, all we need is a left boundary and a right boundary, so we only have to test two values
 - For boundaries of multivariate functions, there could be an infinite number of points on the boundary.
- The *closed interval method* works when the boundary can be reduced to single-variable functions (e.g. line segments), allowing us to determine a finite set of *candidates* for which we can test the function at to find the absolute extrema.
 - However, this method does not work for more complex boundaries (e.g. circles, ellipses, curves, etc.).
- The method of **Lagrange Multipliers** allows us to achieve this step. It is an alternative method of finding a finite set of *candidates* on the boundary.

Method of Lagrange Multipliers with One Constraint

Intuition

- Suppose we want to find the absolute extrema of a function $f(x, y)$ subject to a constraint $g(x, y) = c$ (where c is a constant). The constraint is a level curve of g .
- At the points where f attains its absolute extrema on the constraint, the level curve of f is *tangent* to the level curve of g .
 - This means that the two curves have the same slope at these points.
- The gradients of f and g are orthogonal to their respective level curves. Thus, at the points where f attains its absolute extrema on the constraint, the gradients of f and g are *parallel*.
 - In other words, there exists a scalar λ (called a **Lagrange multiplier**) such that:

$$\vec{\nabla} f(x, y) = \lambda \vec{\nabla} g(x, y).$$

Procedure

1. The first constraint is that the gradients of f and g are parallel: $\vec{\nabla} f(x, y) = \lambda \vec{\nabla} g(x, y)$
This can be rewritten as two equations:

$$f_x = \lambda g_x \qquad f_y = \lambda g_y.$$

2. The second constraint is the original constraint: $g(x, y) = c$.

Together, we have a system of three scalar equations with three unknowns (x, y, λ) :

$$\begin{aligned} f_x &= \lambda g_x \\ f_y &= \lambda g_y \\ g(x, y) &= c. \end{aligned}$$

3. Solving this system gives us a finite number of *candidates* (x, y) for which we can evaluate $f(x, y)$ to find the absolute extrema subject to the constraint.

Method of Lagrange Multipliers with One Constraint

The *candidates* for the absolute extrema of a function $f(x, y)$ subject to the constraint $g(x, y) = c$ occur for all (x, y, λ) that satisfy the system of equations:

$$\vec{\nabla} f = \lambda \vec{\nabla} g$$

$$g(x, y) = c.$$

For a solution $(x, y, \dots, \lambda)$, λ is called the **Lagrange multiplier** for the candidate $(x, y, \dots)$.

Example: Lagrange Multipliers with One Constraint

Find the absolute extrema of $f(x, y) = x^2 + y^2$ subject to the constraint $x + y = 1$.

First, compute the gradients of f and $g(x, y) = x + y$:

$$\vec{\nabla} f(x, y) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle = \langle 2x, 2y \rangle \quad \vec{\nabla} g(x, y) = \left\langle \frac{\partial g}{\partial x}, \frac{\partial g}{\partial y} \right\rangle = \langle 1, 1 \rangle.$$

Next, set up the system of equations:

$$2x = \lambda \quad 2y = \lambda \quad x + y = 1.$$

Solving this system gives two candidates: $(0, 1)$ and $(1, 0)$.

Finally, evaluate f at each candidate:

(x, y)	$f(x, y)$
$(0, 1)$	1
$(1, 0)$	1

Thus, the absolute minimum is 1 at both $(0, 1)$ and $(1, 0)$. There is no absolute maximum since f is unbounded on the constraint.

We can combine this with the standard candidates test from the previous section to find absolute extrema on closed and bounded regions with more complex boundaries:

1. Find the critical points in the interior of the region by setting $\vec{\nabla} f = \vec{0}$ or where $\vec{\nabla} f$ is undefined.
2. Suppose the boundary of f is the constraint $g(x, y) = c$. Then, we can use the method of Lagrange multipliers to find candidates on the boundary.
3. Evaluate f at all candidates from the interior and boundary, and determine the absolute extrema.

Method of Lagrange Multipliers with Multiple Constraints

The method of Lagrange multipliers can be extended to work with multiple constraints.

1. Suppose we want to find the absolute extrema of a function $f(x, y, z)$ subject to two constraints: $g(x, y, z) = c_1$ and $h(x, y, z) = c_2$ (where c_1 and c_2 are constants).

- Each constraint reduces the dimensionality of the domain by one. Thus, with two constraints, we are restricting f to a curve in 3D space. (Whereas with only one, we were restricting f to a surface in 3D space.)
 - The constraint can also be thought of as the *curve of intersection* between the two surfaces defined by $g(x, y, z) = c_1$ and $h(x, y, z) = c_2$.
2. At the points where f attains its absolute extrema on the constraints, the level surface of f is tangent to both constraint surfaces.
- This means that the gradients of f , g , and h are all parallel.
- In other words, there exists scalars λ and μ such that:

$$\vec{\nabla} f(x, y, z) = \lambda \vec{\nabla} g(x, y, z) + \mu \vec{\nabla} h(x, y, z).$$

3. The second and third constraints are the original constraints: $g(x, y, z) = c_1$ and $h(x, y, z) = c_2$. Together, we have a system of five scalar equations with five unknowns (x, y, z, λ, μ) :

$$f_x = \lambda g_x + \mu h_x$$

$$f_y = \lambda g_y + \mu h_y$$

$$f_z = \lambda g_z + \mu h_z$$

$$g(x, y, z) = c_1$$

$$h(x, y, z) = c_2.$$

Method of Lagrange Multipliers with Multiple Constraints

The *candidates* for the absolute extrema of a function $f(x, y, z)$ subject to the constraints $g(x, y, z) = c_1$ and $h(x, y, z) = c_2$ occur for all (x, y, z, λ, μ) that satisfy the system of equations:

$$\vec{\nabla} f = \lambda \vec{\nabla} g + \mu \vec{\nabla} h$$

$$g(x, y, z) = c_1$$

$$h(x, y, z) = c_2.$$

For a solution (x, y, z, λ, μ) , λ and μ are called the **Lagrange multipliers** for (x, y, z) .

Example: Absolute Extrema of a Function of Three Variables

Let $f(x, y, z) = x^2 + 2y^2 - 3xy + z^2$. The closed curve C is defined by the intersection of the surfaces defined by $x^2 + y^2 + z^2 = 6$ and $x + y + z = 0$. The region D is the set of all points on and inside the curve C . Find the absolute extrema of f on the region D .

First, find the critical points in the interior of the region by setting $\vec{\nabla}f = \vec{0}$.

$$\vec{\nabla}f(x, y, z) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle = \langle 2x - 3y, 4y - 3x, 2z \rangle.$$

Setting each component equal to zero gives:

$$\begin{aligned} 2x - 3y &= 0 & 4y - 3x &= 0 & 2z &= 0 \\ (x, y, z) &= (0, 0, 0). \end{aligned}$$

Thus, the only critical point in the interior is $(0, 0, 0)$. Next, use the method of Lagrange multipliers to find candidates on the boundary defined by the constraints $g(x, y, z) = x^2 + y^2 + z^2 = 6$ and $h(x, y, z) = x + y + z = 0$. The gradients of f , g , and h are:

$$\begin{aligned} \vec{\nabla}f(x, y, z) &= \langle 2x - 3y, 4y - 3x, 2z \rangle \\ \vec{\nabla}g(x, y, z) &= \langle 2x, 2y, 2z \rangle & \vec{\nabla}h(x, y, z) &= \langle 1, 1, 1 \rangle. \end{aligned}$$

Setting up the system of equations gives:

$$\begin{aligned} 2x - 3y &= 2x\lambda + \mu \\ 4y - 3x &= 2y\lambda + \mu \\ 2z &= 2z\lambda + \mu \\ x^2 + y^2 + z^2 &= 6 \\ x + y + z &= 0. \end{aligned}$$

Solving this system gives the following candidates on the boundary:

$$\begin{aligned} &(\sqrt{6}, -\sqrt{6}, 0), (-\sqrt{6}, \sqrt{6}, 0), (\sqrt{3}, \sqrt{3}, -2\sqrt{3}), \\ &(-\sqrt{3}, -\sqrt{3}, 2\sqrt{3}), (0, \sqrt{6}, -\sqrt{6}), (0, -\sqrt{6}, \sqrt{6}). \end{aligned}$$

Finally, evaluate f at all candidates:

(x, y, z)	$f(x, y, z)$
$(0, 0, 0)$	0
$(\sqrt{6}, -\sqrt{6}, 0)$ and $(-\sqrt{6}, \sqrt{6}, 0)$	12
$(\sqrt{3}, \sqrt{3}, -2\sqrt{3})$ and $(-\sqrt{3}, -\sqrt{3}, 2\sqrt{3})$	-3
$(0, \sqrt{6}, -\sqrt{6})$ and $(0, -\sqrt{6}, \sqrt{6})$	6

Thus, the absolute minimum is -3 at both $(\sqrt{3}, \sqrt{3}, -2\sqrt{3})$ and $(-\sqrt{3}, -\sqrt{3}, 2\sqrt{3})$. The absolute maximum is 12 at both $(\sqrt{6}, -\sqrt{6}, 0)$ and $(-\sqrt{6}, \sqrt{6}, 0)$.

4 Multivariate Integrals

4.1 Double Integrals

- **Recall:** If we have a function $f(x)$, we can find the area “under the curve” over the interval $[a, b]$ by computing the integral $\int_a^b f(x) dx$.
- A **double integral** can be used to find the *volume* “under the surface” defined by $z = f(x, y)$.

Riemann Sums to Approximate a Double Integral

Recall: The *Riemann sum* used to approximate the area under $f(x)$ from $x = a$ to $x = b$ is:

$$A \approx \sum_{k=0}^n f(x_k) \Delta x$$

where $\Delta x = \frac{b-a}{n}$, the width of each subinterval, and x_k is a “sample point” in the k^{th} subinterval.

Riemann Sum for Two Variables

1. Define a region and partition

Let R be a rectangular region in the xy -plane defined by $R = [a, b] \times [c, d]$. We can partition R into m subintervals along the x -axis and n subintervals along the y -axis, creating a grid of rectangles.

If we use equal subintervals, we can call $\Delta x = \frac{b-a}{m}$ and $\Delta y = \frac{d-c}{n}$.

2. Define sample points

For each subrectangle, we can choose a sample point (x_i, y_j) within that subrectangle. Then the approximate height to the surface is given by $z_{ij} = f(x_i, y_j)$.

For example, we could choose the sample point to be the lower-left corner of each subrectangle, meaning $x_i = a + i\Delta x$ and $y_j = c + j\Delta y$.

3. Define volume of each box

The area of each subrectangle is given by $\Delta A = \Delta x \Delta y$.

The volume of the box approximating the volume under the surface over each subrectangle is thus $f(x_i, y_j) \Delta A = f(x_i, y_j) \Delta x \Delta y$.

4. Definition of Riemann sum

The **Riemann sum** approximating the volume under the surface over the region R is given by summing the volumes of all the boxes:

$$V \approx \sum_{i=0}^m \sum_{j=0}^n f(x_i, y_j) \Delta A = \sum_{i=0}^m \sum_{j=0}^n f(x_i, y_j) \Delta x \Delta y.$$

The definite integral of $f(x)$ from $x = a$ to $x = b$ (the actual area) is when $n \rightarrow \infty$:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=0}^n f(x_k) \Delta x.$$

Limit Definition of a Double Integral

The double integral of $f(x, y)$ over the rectangular region $R = [a, b] \times [c, d]$ is defined as:

$$\iint_R f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{i=0}^m \sum_{j=0}^n f(x_i, y_j) \Delta A$$

where $\Delta A = \Delta x \Delta y$ is the area of each subrectangle, and (x_i, y_j) is a sample point in each subrectangle.

It is simply when we take infinitely many partitions using the Riemann sum.

Example: Riemann Sum for Volume

Use a Riemann Sum to approximate the volume under the surface $z = 9xe^{-xy}$ over the rectangular region $R = [0, 2] \times [0, 1]$ using 2 subintervals in each direction and choosing the sample points to be the upper-right corners of each subrectangle.

The upper-right corners of each subrectangle are given by:

- For $i = 1, j = 1$: $(x_1, y_1) = (1, 0.5)$
- For $i = 1, j = 2$: $(x_1, y_2) = (1, 1)$
- For $i = 2, j = 1$: $(x_2, y_1) = (2, 0.5)$
- For $i = 2, j = 2$: $(x_2, y_2) = (2, 1)$

4.2 Iterative Integration

Fubini's Theorem

If $f(x, y)$ is continuous on the rectangle $R = [a, b] \times [c, d]$, then the double integral of f over R can be computed as an iterated integral in either order:

$$\iint_R f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy$$

Similarly, if we can write $f(x, y)$ as a product of functions of x and y , then we can separate the double integral:

$$\int_a^b \int_c^d g(x)h(y) dy dx = \int_a^b g(x) dx \int_c^d h(y) dy.$$

Partial Integration

- The indefinite integral of a function with multiple variables can be computed by integrating with respect to one variable at a time, treating the other variables as constants. This is known as **partial integration**.
- For example, to compute the partial integral of $f(x, y)$ with respect to x :

$$\int f(x, y) dx = F(x, y) + C(y)$$

where $C(y)$ is an arbitrary function of y .

- Similarly, to compute the partial integral of $f(x, y)$ with respect to y :

$$\int f(x, y) dy = G(x, y) + D(x)$$

where $D(x)$ is an arbitrary function of x .

4.3 Double Integrals as Volume

Double Integrals over Rectangular Regions

- The volume under the surface $z = f(x, y)$ over a rectangular region $R = [a, b] \times [c, d]$ is given by the double integral:

$$V = \int_R f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx$$

Double Integrals over General Regions

Double Integrals in Polar Coordinates

4.4 Applications of Double Integrals

Surface Area

4.5 Triple Integrals

4.6 Interpretations of Triple Integrals

Triple Integrals in Cylindrical Coordinates

Triple Integrals in Spherical Coordinates

4.7 The Jacobian

5 Vector Calculus

5.1 Vector Fields

- A **vector field** assigns a vector to every point in space.
 - In other words, it maps $\mathbb{R}^n \rightarrow \mathbb{R}^n$. It is a function that takes in n inputs and outputs a vector with n components.

- A vector field in two-dimensions ($\mathbb{R}^2$) can be represented as:

$$\vec{F}(x, y) = \langle P(x, y), Q(x, y) \rangle.$$

Here, P and Q are called the **component functions** of $\vec{F}$.

- Similarly, a vector field in three-dimensions ($\mathbb{R}^3$) can be represented as:

$$\vec{F}(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z) \rangle.$$

- In a general sense, a “field” assigns quantities to every point in space.
 - We can generalize this by saying a **scalar field** assigns a scalar to every point in space. It is a function $\mathbb{R}^n \rightarrow \mathbb{R}$ that takes in n inputs and outputs a single scalar value.
- Examples of where vector fields are used:
 - Fluid flow: the velocity of a fluid at different points in space
 - Electromagnetism: electric and magnetic fields
 - Force fields: the force at different points in space (such as gravitational)
 - **Gradient fields**: the gradient of a scalar function is a vector field

Introduction to Conservative Vector Fields

A recurring theme in vector calculus is the idea of **conservative vector fields**. These are vector fields that can be represented as the gradient of some scalar function. Conservative vector fields have special properties that make them easier to work with.

Formally, a vector field $\vec{F}$ is **conservative** if there exists a scalar function f such that $\vec{F} = \vec{\nabla}f$. The function f is called a **potential function** for the vector field.

We will explore conservative vector fields further beginning in Section 5.3.

Example: Is a Vector Field Conservative?

Show that the vector field $\vec{F}(x, y) =$

5.2 Line Integrals

- Line integrals allow us to integrate along any curve in space, rather than just along one axis.
 - Instead of integrating over an interval $[a, b]$, we integrate along a curve C .
 - Moving along C can involve changes in multiple dimensions at once.
- A line integral is

Line Integral of Scalar Fields

The line integral of a scalar field $f(\vec{r}(t))$ from $t = a$ to $t = b$ along a curve C parameterized by $\vec{r}(t)$ is given by:

$$\int_C f(\vec{r}) ds = \int_a^b f(\vec{r}(t)) |\vec{r}'(t)| dt.$$

- Examples of where line integrals are used:
 - Calculating work done by a force field along a path
 - The mass of a wire with varying density.

Example: Evaluating a Line Integral

Evaluate the line integral $\int_C (2 + x^2 y) ds$, where C is the upper half of the circle $x^2 + y^2 = 1$ from $(1, 0)$ to $(-1, 0)$.

First, parameterize the curve C . A suitable parameterization is:

$$\vec{r}(t) = \langle \cos t, \sin t \rangle \quad 0 \leq t \leq \pi.$$

Use this parameterization to find $ds = |\vec{r}'(t)| dt$:

$$\begin{aligned} \vec{r}'(t) &= \langle -\sin t, \cos t \rangle \\ ds &= |\vec{r}'(t)| dt = \sqrt{(-\sin t)^2 + (\cos t)^2} = \sqrt{\sin^2 t + \cos^2 t} dt = 1 dt. \end{aligned}$$

Plug in $x = \cos t$ and $y = \sin t$ from the parameterization into the integrand:

$$2 + x^2 y = 2 + (\cos t)^2 \sin t.$$

Thus, the line integral becomes:

$$\int_C (2 + x^2 y) ds = \int_0^\pi (2 + (\cos t)^2 \sin t) 1 dt = \int_0^\pi (2 + \cos^2 t \sin t) dt.$$

Evaluate the integral:

$$\int_0^\pi (2 + \cos^2 t \sin t) dt = \left(2t - \frac{\cos^3 t}{3} \right) \Big|_0^\pi = \left(2\pi + \frac{1}{3} \right) + \frac{1}{3} = 2\pi + \frac{2}{3}.$$

Classifying Curves

It is important to classify curves correctly when working with line integrals, since some properties and theorems of line integrals are only valid for line integrals over certain types of curves.

- A curve is **simple** if it does not cross itself.
- A curve is **closed** if it starts and ends at the same point.
 - A closed curve divides the plane into an “interior” and “exterior” region.
- A curve is **positively oriented** if it moves counterclockwise around its interior region.
 - If you followed the curve, its interior should be to your left.
 - A curve that moves clockwise is **negatively oriented**.
- A curve is **smooth** if its parameterization $\vec{r}(t)$ is differentiable (and $\vec{r}'(t) \neq \vec{0}$ for all t).
 - A curve is **piecewise-smooth** if it can be broken into a finite number of smooth segments.

Classifying Regions

- A region is **open** if we do not include its boundary. (think of a $<$ inequality instead of $\leq$)
 - Otherwise, the region is **closed**.
- A region is **connected** if any two points in the region can be connected by a path that lies entirely within the region. (Think connected graph)
- A region is **simply connected** if it is connected and has no holes.
 - In other words, any closed curve within the region can be continuously shrunk to a point without leaving the region.

Line Integrals of Vector Fields

When moving along a curve, we originally defined infinitesimal movements along the curve as $ds = \sqrt{dx^2 + dy^2}$ (in 2D at least). This was suitable for line integrals of scalar fields.

Integrating over a vector field means we must somehow sum up the product of the vector field and the movement along the curve as we move along it. There are many ways we can do this:

1. Continue using ds : $\int_C \vec{F} ds$. This gives us a vector result.
2. Use the component of the vector field aligned with the direction of movement along the curve (the tangent vector): $\int_C \vec{F} \cdot d\vec{r}$. This gives us a scalar result.

These are all line integrals over vector fields, but the one which has the most applicability is the second one, which is interpreted as the work done by the vector field along the curve.

When moving along a curve C parameterized by $\vec{r}(t)$, the infinitesimal displacement vector is given by $d\vec{r} = \langle dx, dy \rangle$. We find that $d\vec{r} = \vec{r}'(t) dt = \langle x'(t), y'(t) \rangle dt$ by the fundamental theorem.

Line Integral of Vector Fields

The line integral of a vector field $\vec{F}$ from $t = a$ to $t = b$ along a curve C parameterized by $\vec{r}(t)$ is given by:

$$\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt.$$

In the case of $\mathbb{R}^2$, if $\vec{F} = \langle P, Q \rangle$, then we can also write out the dot product:

$$\int_C \vec{F} \cdot d\vec{r} = \int_C P dx + Q dy.$$

We can turn a line integral of a vector field into a normal integral by realizing that P is a function of $x(t)$ and $y(t)$ and dx and dy can be expanded to be in terms of t :

$$\int_C P dx + Q dy = \int_a^b \left(P(x(t), y(t))x'(t) dt + Q(x(t), y(t))y'(t) \right) dt.$$

Example: Evaluating Line Integral of a Vector Field

Example: Work Done by a Force Field

5.3 Fundamental Theorem of Line Integrals

Recall the **Fundamental Theorem of Calculus**, which says that the integral of a function's derivative is the difference of the function's values at the endpoints:

$$\int_a^b f'(x) dx = f(b) - f(a)$$

We can generalize this theorem by noticing that:

- the *integral* is the *sum of little changes* from a to b
 - each *little change* in f is the differential $df = f'(x) dx$
- $f(b) - f(a)$ is the *total change* in f from a to b , which we can call Δf
- so, the sum of little changes in f from a to b is equal to the total change in f from a to b :

$$\int_a^b df = \Delta f = f(b) - f(a).$$

The fundamental theorem would imply that given some function f , the sum of little changes in f over some curve is equal to the total change in f from the start of the curve to the end.

We know the total differential of $f(x, y, z)$ is $df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$. Each total differential df can be interpreted as a “little change in f ”. If $\vec{p}$ and $\vec{q}$ are the start and end points of some curve C , then by the fundamental theorem, we can say:

$$\int_C df = \Delta f = f(\vec{q}) - f(\vec{p}).$$

Realize that the $df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$ can be represented as the dot product $\vec{\nabla} f \cdot d\vec{r}$, since $\vec{\nabla} f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$ and $d\vec{r} = \langle dx, dy, dz \rangle$. Thus, this gives us the form:

Fundamental Theorem of Calculus for Line Integrals (Gradient Theorem)

Let C be a curve that starts at displacement $\vec{p}$ and ends at displacement $\vec{q}$, and f is a function such that $\vec{\nabla} f$ is continuous on C . Then:

$$\int_C \vec{\nabla} f \cdot d\vec{r} = f(\vec{q}) - f(\vec{p})$$

for any function $f(\vec{r})$.

We can think of $\vec{\nabla} f$ as the “little changes in f ” and integrating over $\vec{\nabla} f$ over the curve C gives us the net change of f over the boundary of C , which are the endpoints.

If C is parameterized using t , then since $d\vec{r} = \vec{r}'(t) dt$, we can write:

$$\int_a^b \vec{\nabla} f(\vec{r}(t)) \cdot \vec{r}'(t) dt = f(\vec{r}(b)) - f(\vec{r}(a)).$$

Note that if C is piecewise-defined, even though the integral over C is improper, the gradient theorem would hold due to properties of integrals (we can split the integral apart).

Conservative Vector Fields

The gradient theorem tells us that the line integral of $\vec{\nabla}f$ along two different paths is equal, as long as the paths start and end in the same position. In other words, if C_1 and C_2 have the same initial and terminal points, then:

$$\int_{C_1} \vec{\nabla}f \cdot d\vec{r} = \int_{C_2} \vec{\nabla}f \cdot d\vec{r}.$$

In other words, if a vector field $\vec{F}$ can be represented as the gradient of some function f (if there exists an f such that $\vec{F} = \vec{\nabla}f$), then the line integral $\int_C \vec{F} \cdot d\vec{r} = \int_C \vec{\nabla}f \cdot d\vec{r}$ is independent of path and only depends on the initial and terminal points of C .

Vector fields which meet this condition ($\vec{F} = \vec{\nabla}f$) are known as **conservative vector fields**. We call the function f a **potential function** for the vector field $\vec{F}$.

Conservative Vector Field

A vector field $\vec{F}$ is **conservative** if there exists a function f such that $\vec{F} = \vec{\nabla}f$. In this case, f is called a **potential function** for $\vec{F}$.

Applying the Gradient Theorem to Conservative Vector Fields

If $\vec{F}$ is conservative, we can apply the gradient theorem to find line integrals over $\vec{F}$ easily by evaluating the potential function f at the endpoints of the curve.

In $\mathbb{R}^2$, this means if we can define $\vec{F} = \langle P, Q \rangle$, then $\vec{\nabla}f = \vec{F} \implies \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle = \langle P, Q \rangle$. This is a system of equations that we can solve to find f .

Example: Finding a Potential Function

Find a potential function for the conservative vector field $\vec{F}(x, y) = \langle 2xy, x^2 \rangle$.

We need to find f such that $\vec{\nabla}f = \vec{F}$. This gives us the system of equations:

$$\frac{\partial f}{\partial x} = 2xy \qquad \frac{\partial f}{\partial y} = x^2$$

Partial integration of the first equation with respect to x gives:

$$f(x, y) = \int 2xy \, dx = x^2y + g(y).$$

Next, we differentiate this expression for f with respect to y :

$$\frac{\partial f}{\partial y} = x^2 + g'(y).$$

Setting this equal to the second equation from our system, we have:

$$x^2 + g'(y) = x^2 \implies g'(y) = 0.$$

Integrating this gives $g(y) = C$. Thus, a potential function for the vector field is:

$$f(x, y) = x^2y + C.$$

5.4 Green's Theorem

Often, line integrals can be difficult to compute. **Green's Theorem** states that a line integral over a closed path can be evaluated as a double integral over the region enclosed by the path.

Green's Theorem

Let D be the region bounded by the closed curve C and $\vec{F} = \langle P, Q \rangle$. Then:

$$\oint_C \vec{F} \cdot d\vec{r} = \oint_C (P dx + Q dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

if C is simple, closed, and positively-oriented.

Constraints of Green's Theorem

- **Simple closed curves.** Recall that *simple* means the curve does not cross itself. *Closed* means the curve will start and end at the same point.
- **Orientation.** The definition above applies only for *positively oriented* curves, i.e. those that move counterclockwise. If you followed the curve, its interior should be to your left. If the curve is *negatively oriented*, then we must negate the sign of the double integral:

$$\oint_C \vec{F} \cdot d\vec{r} = - \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA \text{ if } C \text{ is negatively oriented.}$$

Example: Evaluating a Line Integral using Green's Theorem

Use Green's Theorem to evaluate the line integral $\oint_C (x^2 - y) dx + (x + y^2) dy$, where C is the positively-oriented triangle with vertices at $(0, 0)$, $(1, 0)$, and $(1, 1)$.

First, we identify $P(x, y) = x^2 - y$ and $Q(x, y) = x + y^2$. Next, we compute the partial derivatives:

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(x + y^2) = 1 \qquad \frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(x^2 - y) = -1.$$

Thus, $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1 - (-1) = 2$.

Next, we need to find the area of the triangular region D bounded by C . The triangle has a base and height of length 1, so its area is $A_D = \frac{1 \cdot 1}{2} = \frac{1}{2}$.

Finally, we can apply Green's Theorem:

$$\oint_C (x^2 - y) dx + (x + y^2) dy = \iint_D 2 dA = 2A_D = 2\left(\frac{1}{2}\right) = 1.$$

Example: Green's Theorem with a Polar Integral

Let $\vec{F} = \langle y^2, 3xy \rangle$. Use Green's Theorem to evaluate the line integral $\oint_C \vec{F} \cdot d\vec{r}$, where C is the positively-oriented circle of radius 2 centered at the origin.

First, identify $P = y^2$ and $Q = 3xy$ and compute the partial derivatives:

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(3xy) = 3y \qquad \frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(y^2) = 2y.$$

Thus, $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 3y - 2y = y$.

Next, set up the polar integral for the right side of Green's Theorem:

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_D y \, dA = \int_0^{2\pi} \int_0^2 \underbrace{r \sin \theta}_{y} \underbrace{r \, dr \, d\theta}_{dA}.$$

We are integrating over the whole circle of radius 2, so θ goes from 0 to 2π and r goes from 0 to 2.

Evaluate the integral:

$$\begin{aligned} \int_0^{2\pi} \int_0^2 r^2 \sin \theta \, dr \, d\theta &= \int_0^{2\pi} \sin \theta \, d\theta \int_0^2 r^2 \, dr \\ &= \left(-\cos \theta \right) \Big|_0^{2\pi} \left(\frac{r^3}{3} \right) \Big|_0^2 \\ &= (-\cos(2\pi) + \cos(0)) \left(\left(\frac{8}{3} \right) - 0 \right) = 0. \end{aligned}$$

Example: Deriving the Area of an Ellipse using Green's Theorem

An ellipse centered at the origin is given by the parametric representation $\vec{r}(t) = \langle a \cos t, b \sin t \rangle$ for $0 \leq t < 2\pi$. Use Green's Theorem to find the area of the ellipse.

If we want the right-hand side of Green's Theorem to equal the area of the enclosed region, we want $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$. Any functions P and Q that satisfy this will work.

Let's choose $P = 0$ and $Q = x$. This gives $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1 - 0 = 1$.

Thus, by Green's Theorem:

$$\oint_C 0 \, dx + x \, dy = \iint_D 1 \, dA = A_{\text{ellipse}} \implies A_{\text{ellipse}} = \oint_C x \, dy.$$

We know $x = a \cos t$ and $y = b \sin t \implies dy = b \cos t \, dt$. Thus:

$$A_{\text{ellipse}} = \int_0^{2\pi} a \cos t (b \cos t) \, dt = ab \int_0^{2\pi} \cos^2 t \, dt.$$

Using the identity $\cos^2 t = \frac{1 + \cos(2t)}{2}$, we have:

$$A_{\text{ellipse}} = \frac{ab}{2} \int_0^{2\pi} (1 + \cos(2t)) \, dt = \frac{ab}{2} \left[t + \left(\frac{1}{2} \right) \sin(2t) \right]_0^{2\pi} = \frac{ab}{2} (2\pi - 0) = \pi ab.$$

5.5 Divergence and Curl

Scalar fields have gradients to describe how they change in space. Vector fields have two analogous operations to describe how they change in space: **divergence** and **curl**.

Divergence of a Vector Field

The divergence of a vector field $\vec{F}(x, y, z) = \langle P, Q, R \rangle$ is:

$$\operatorname{div} \vec{F} = \vec{\nabla} \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}.$$

The notation $\vec{\nabla} \cdot \vec{F}$ can be used to represent divergence, treating $\vec{\nabla}$ as a vector of partial derivative operators:

$$\vec{\nabla} \cdot \vec{F} = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle \cdot \langle P, Q, R \rangle = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}.$$

We will use this definition of $\vec{\nabla}$ to derive the *curl* of a vector field:

Curl of a Vector Field

The curl of a vector field $\vec{F}(x, y, z) = \langle P, Q, R \rangle$ is:

$$\operatorname{curl} \vec{F} = \vec{\nabla} \times \vec{F} = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle \times \langle P, Q, R \rangle.$$

By the determinant definition of the cross product, this gives:

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = \begin{pmatrix} \frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \\ \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \\ \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \end{pmatrix}$$

Example: Finding Divergence and Curl

Find the divergence and curl of the vector field $\vec{F}(x, y, z) = \langle x^2y, y^2z, z^2x \rangle$.

The divergence is:

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial x^2y}{\partial x} + \frac{\partial y^2z}{\partial y} + \frac{\partial z^2x}{\partial z} = 2xy + 2yz + 2zx.$$

The curl is:

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y & y^2z & z^2x \end{vmatrix} = \langle -2yz, -2xz, 2xy \rangle.$$

Properties of Divergence and Curl

- The divergence of a vector field is scalar, while the curl of a vector field is a vector.
- The divergence and curl operators are linear:
 - $\text{div}(a\vec{F} + b\vec{G}) = a \text{div } \vec{F} + b \text{div } \vec{G}$
 - $\text{curl}(a\vec{F} + b\vec{G}) = a \text{curl } \vec{F} + b \text{curl } \vec{G}$
- **The curl of a conservative vector field is zero.** (see derivation below)
 - If $\vec{F} = \vec{\nabla}f$, then $\text{curl } \vec{F} = \vec{\nabla} \times \vec{\nabla}f = \vec{0}$.

Proof: Conservative vector fields have zero curl

Let $\vec{F} = \vec{\nabla}f$ be a conservative vector field. Then, by the definition of curl:

$$\text{curl } \vec{F} = \vec{\nabla} \times \vec{F} = \vec{\nabla} \times (\vec{\nabla}f) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix}.$$

Evaluating this determinant gives:

$$\vec{\nabla} \times (\vec{\nabla}f) = \left\langle \frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y}, \frac{\partial^2 f}{\partial z \partial x} - \frac{\partial^2 f}{\partial x \partial z}, \frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x} \right\rangle.$$

By Clairaut's Theorem, the mixed partial derivatives are equal, so each component of this vector is zero. Thus:

$$\text{curl } \vec{F} = \langle 0, 0, 0 \rangle = \vec{0}.$$

2D Curl

In two dimensions, we can treat a vector field $\vec{F}(x, y) = \langle P, Q \rangle$ as a three-dimensional vector field with the third component equal to zero: $\vec{F}(x, y, z) = \langle P, Q, 0 \rangle$. The curl of this is:

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & 0 \end{vmatrix} = \left\langle 0, 0, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right\rangle.$$

The magnitude of this curl vector is simply $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$, which is a scalar. Thus, in two dimensions, we can define the curl as a scalar quantity:

$$\text{2d-curl } \vec{F} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}.$$

- In 2D, vector fields can only curl along one plane (the plane of the field itself), so there is no need to represent the curl as a vector.
-

Interpreting Divergence and Curl

- **Divergence** measures how much a vector field is “spreading out” or “converging” at a point.
 - A positive divergence indicates a “source” where vectors are spreading out (“diverging”).
 - A negative divergence indicates a “sink” where vectors are converging.

- A vector field with zero divergence is called a **solenoidal** or **incompressible** vector field.
- **Curl** measures the tendency of a vector field to rotate around a point.
 - A non-zero curl indicates rotational behavior in the field.
 - The direction of the curl vector indicates the axis of rotation, following the right-hand rule.
 - A vector field with zero curl is called an **irrotational** vector field.

Extension: Laplacian Operator

The **Laplacian** of a scalar function f , $\nabla^2 f$, is the divergence of its gradient:

$$\nabla^2 f = \vec{\nabla} \cdot \vec{\nabla} f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}.$$

A function f is called **harmonic** if its Laplacian is zero:

$$\nabla^2 f = 0 \implies f \text{ is harmonic.}$$

5.6 Surface and Flux Integrals

Surface Integrals

Recall that we can take line integrals over scalar fields by integrating along little pieces of length ds along a curve C . These pieces ds represent little changes in displacement along the curve, and we derived that $ds = |\vec{r}'(t)| dt$ in the arc length unit.

Similarly, we can take **surface integrals** over scalar fields by summing up little pieces of surface area dS over a surface S . In the surface area unit, we derived the two formulas for dS :

- For a parametric surface $\vec{r}(u, v)$: $dS = |\vec{r}_u \times \vec{r}_v| du dv$.
- For a surface defined as $z = g(x, y)$: $dS = \sqrt{1 + (g_x)^2 + (g_y)^2} dA$.

Surface Integral

The surface integral of a scalar field $f(\vec{r}(u, v))$ over a parametric surface S parameterized by $\vec{r}(u, v)$ is given by:

$$\iint_S f(\vec{r}(u, v)) dS = \iint_D f(\vec{r}(u, v)) |\vec{r}_u \times \vec{r}_v| du dv.$$

where D is the domain of the parameters u and v .

If we let $\vec{r}(x, y) = \langle x, y, g(x, y) \rangle$ which defines the surface $z = g(x, y)$, we find that:

$$\iint_S f(x, y, z) dS = \iint_D f(x, y, g(x, y)) \sqrt{1 + (g_x)^2 + (g_y)^2} dA.$$

Example: Surface Integral over a Parametric Surface

Find $\iint_S xyz \, dS$ if S is the cone $\vec{r}(u, v) = \langle u \cos v, u \sin v, u \rangle$ ($0 \leq u \leq 3$ and $0 \leq v \leq \frac{\pi}{2}$).

First, compute $dS = |\vec{r}_u \times \vec{r}_v| \, du \, dv$:

$$\begin{aligned}\vec{r}_u &= \langle \cos v, \sin v, 1 \rangle & \vec{r}_v &= \langle -u \sin v, u \cos v, 0 \rangle \\ \vec{r}_u \times \vec{r}_v &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos v & \sin v & 1 \\ -u \sin v & u \cos v & 0 \end{vmatrix} = \langle -u \cos v, -u \sin v, u \rangle\end{aligned}$$

$$\begin{aligned}dS &= |\vec{r}_u \times \vec{r}_v| \, du \, dv = \sqrt{(-u \cos v)^2 + (-u \sin v)^2 + u^2} \, du \, dv \\ &= \sqrt{u^2(\cos^2 v + \sin^2 v + 1)} \, du \, dv = \sqrt{2u^2} = \sqrt{2}u \, du \, dv.\end{aligned}$$

Next, plug in the parameterization into the integrand:

$$xyz = (u \cos v)(u \sin v)(u) = u^3 \cos v \sin v.$$

Thus, the surface integral becomes:

$$\iint_S xyz \, dS = \int_0^{\frac{\pi}{2}} \int_0^3 (u^3 \cos v \sin v) (\sqrt{2}u \, du \, dv) = \sqrt{2} \int_0^{\frac{\pi}{2}} \cos v \sin v \, dv \int_0^3 u^4 \, du.$$

Evaluate the integral:

$$\sqrt{2} \frac{\sin^2 v}{2} \bigg|_0^{\frac{\pi}{2}} \frac{u^5}{5} \bigg|_0^3 = \sqrt{2} \left(\frac{1}{2} - 0 \right) \left(\frac{243}{5} - 0 \right) = \frac{243\sqrt{2}}{10}.$$

Example: Surface Integral over a Function-Defined Surface

Flux Integrals

Just like how we were able to take line integrals not only over scalar fields but also over vector fields, we can do the same with surface integrals.

- For a line integral, we chose to take the component of each vector field aligned with the direction it is moving along the curve (the tangent vector).
- For a surface integral, we instead take the component of each vector field aligned with the normal vector to the tangent plane at that point. This has the effect of measuring how much of the vector field is “flowing through” the surface, rather than along it, so it is often called a **flux integral**.

Flux in 3D

The flux of a vector field $\vec{F}$ through a surface S is given by the surface integral:

$$\text{flux} = \Phi = \iint_S \vec{F} \cdot \hat{n} \, dS = \iint_S \vec{F} \cdot d\vec{S}.$$

Where $\hat{n}$ is the unit normal vector to the surface at each point. Note that $d\vec{S} = \hat{n} \, dS$.

Note that each tangent plane has two possible normal vectors (pointing in opposite directions). Thus, when computing flux integrals, it is important to specify the orientation of the surface (which normal vector to use).

- A **positively oriented** surface is one where the normal vectors point outward from the enclosed volume.
- A **negatively oriented** surface is one where the normal vectors point inward toward the enclosed volume.

Classifying Surfaces of Integration

- A surface is **smooth** if it can be parameterized by a differentiable function $\vec{r}(u, v)$ with continuous partial derivatives (and $\vec{r}_u \times \vec{r}_v \neq \vec{0}$ for all (u, v)).
 - A surface is **piecewise-smooth** if it can be broken into a finite number of smooth segments.
- A surface is **orientable** if we can consistently define a normal vector at every point on the surface.
 - Intuitively, this means we can choose one side of the surface to be the “top” side and the other to be the “bottom” side without any contradictions (e.g., a Mobius strip is not orientable)

If S is parametrically defined using $\vec{r}(u, v)$, then recall that the normal vector to the surface is given by $\vec{r}_u \times \vec{r}_v$. The unit normal vector is then:

$$\hat{n} = \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|}.$$

Plugging this into the flux integral definition gives:

$$\iint_S \vec{F} \cdot \hat{n} \, dS = \iint_S \vec{F} \cdot \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} |\vec{r}_u \times \vec{r}_v| \, du \, dv = \iint_D \vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) \, du \, dv.$$

Flux Integral over a Parametric Surface

The flux Φ of a vector field $\vec{F}$ through a positively oriented parametric surface S parameterized by $\vec{r}(u, v)$ is:

$$\Phi = \iint_D \vec{F} \cdot (\vec{r}_u \times \vec{r}_v) \, du \, dv.$$

If S is negatively oriented, then use $\vec{r}_v \times \vec{r}_u$ instead.

For a surface defined as $z = g(x, y)$, the parametric representation is $\vec{r}(x, y) = \langle x, y, g(x, y) \rangle$, and $\vec{r}_x \times \vec{r}_y = \langle -g_x, -g_y, 1 \rangle$. Thus, the flux integral becomes:

$$\Phi = \iint_S \vec{F} \cdot (\vec{r}_x \times \vec{r}_y) dA = \iint_D \vec{F}(x, y, g(x, y)) \cdot \langle -g_x, -g_y, 1 \rangle dA.$$

If we let $\vec{F} = \langle P, Q, R \rangle$, then computing the dot product gives:

$$\Phi = \iint_D (-Pg_x - Qg_y + R) dA.$$

Note that this assumes the surface is positively oriented. If the surface is negatively oriented, we must negate the flux integral:

$$\Phi = - \iint_D (-Pg_x - Qg_y + R) dA = \iint_D (Pg_x + Qg_y - R) dA.$$

Flux Integral over a Function-Defined Surface

The flux Φ of a vector field $\vec{F} = \langle P, Q, R \rangle$ through a positively oriented surface S defined by $z = g(x, y)$ above a region D is:

$$\Phi = \iint_S \vec{F} \cdot d\vec{S} = \iint_D (-Pg_x - Qg_y + R) dA.$$

If S is negatively oriented, then: $\Phi = \iint_D (Pg_x + Qg_y - R) dA$.

Example: Computing a Flux Integral

Flux in 2D

We can generalize flux to two dimensions as well.

When taking a line integral over a curve in 2D, we used to take the component of the vector field aligned with the tangent vector: $\int_C \vec{F} \cdot \vec{r}'(t) dt$. What if instead, we took the component of the vector field aligned with the normal vector to the curve at each point? This is flux.

In 2D, flux measures how much of a vector field *passes through* a curve C . The flux through a curve is given by the line integral:

$$\text{flux} = \Phi = \int_C \vec{F} \cdot \hat{n} ds.$$

Again, when computing flux integrals, it is important to specify the orientation of the curve (which normal vector to use).

We can use $\hat{n} = \vec{N}$, the unit normal vector to the curve at each point (recall from unit 2 that the unit normal vector is found by first finding the unit tangent vector $\vec{T} = \frac{\vec{r}'}{|\vec{r}'|}$ and then taking $\vec{N} = \frac{\vec{T}'}{|\vec{T}'|}$). $\vec{N}$ points towards the interior of curvature C , which is a negative orientation.

Thus, we should use $\hat{n} = -\vec{N}$ if we desire a positive orientation (“outwards flux”):

$$\Phi = - \int_C \vec{F} \cdot \vec{N} \, ds.$$

Derivation: 2D Flux Integral Formula

Let $\vec{F} = \langle P, Q \rangle$ and $\vec{N}(t)$ be the unit normal vector to the curve C parameterized by $\vec{r}(t)$.

Also, $ds = |\vec{r}'(t)| \, dt = \langle dx, dy \rangle \implies \frac{ds}{dt} = |\vec{r}'(t)|$ by definition.

$$\begin{aligned} \vec{T} &= \frac{\vec{r}'}{|\vec{r}'|} = \frac{\left\langle \frac{dx}{dt}, \frac{dy}{dt} \right\rangle}{ds/dt} = \left\langle \frac{dx}{ds}, \frac{dy}{ds} \right\rangle \\ \vec{T}' &= \left\langle \frac{d}{dt} \frac{dx}{ds}, \frac{d}{dt} \frac{dy}{ds} \right\rangle = \frac{dt}{ds} \left\langle \frac{d^2x}{ds^2}, \frac{d^2y}{ds^2} \right\rangle = \frac{1}{ds/dt} \left\langle \frac{d^2x}{ds^2}, \frac{d^2y}{ds^2} \right\rangle \\ |\vec{T}'| &= \frac{1}{ds/dt} \\ \vec{N} &= \frac{\vec{T}'}{|\vec{T}'|} = \end{aligned}$$

Flux Integral in 2D

The flux Φ of a vector field $\vec{F}$ through a closed curve C in $\mathbb{R}^2$ is given by the line integral:

$$\Phi = \int_C \vec{F} \cdot \hat{n} \, ds = \int_C P \, dy - Q \, dx.$$

Green's Theorem for Flux

Flux-Divergence Form of Green's Theorem

Let D be the region bounded by the closed curve C and $\vec{F} = \langle P, Q \rangle$. Then:

$$\oint_C \vec{F} \cdot \vec{N} \, ds = \oint_C P \, dy - Q \, dx = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dA.$$

5.7 Stokes' Theorem

Stokes' Theorem (Curl Theorem)

Let C be a closed curve that encloses a surface D . Then Stokes' Theorem states that:

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_D (\text{curl } \vec{F}) \cdot \hat{n} dS.$$

where C is **simple, closed, and positively-oriented**, and $\hat{n}$ is the unit normal vector to the surface D .

In other words, the circulation of a vector field around a curve is equal to adding all the little curls over the surface enclosed by that curve.

Notice that if use 2D curl instead of 3D curl in Stokes' Theorem, we get Green's Theorem. Thus, Green's Theorem can be interpreted as a special case of Stokes' Theorem in two dimensions:

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_D (2\text{d-curl } \vec{F}) dA = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

Constraints of Stokes' Theorem

- **Simple closed curves.** Recall that *simple* means the curve does not cross itself. *Closed* means the curve will start and end at the same point.
- **Orientation.** The definition above applies only for *positively oriented* curves, i.e. those that move counterclockwise. If you followed the curve, its interior should be to your left. If the curve is *negatively oriented*, then we must negate the sign of the surface integral:

$$\oint_C \vec{F} \cdot d\vec{r} = - \iint_D (\text{curl } \vec{F}) \cdot \hat{n} dS$$

if C is **simple, closed, and negatively-oriented**.

Example: Using Stokes' Theorem to Evaluate a Line Integral

Example: Using Stokes' Theorem to Evaluate a Surface Integral

5.8 Divergence Theorem

Divergence Theorem (Gauss' Theorem)

Let S be a positively-oriented, closed surface that encloses a solid region V . Then:

$$\oiint_S \vec{F} \cdot \hat{n} dS = \iiint_V (\text{div } \vec{F}) dV.$$

In simpler terms, the total flux out of the volume's boundary is the result of adding up all the little "divergences" inside the volume.

Constraints of the Divergence Theorem

- **Closed surface.** The surface S must be closed, meaning it completely encloses a volume without any gaps or holes.
- **Orientation.** The surface S must be positively oriented, meaning the normal vectors point outward from the enclosed volume.
- **Continuity.** The vector field $\vec{F}$ must have continuous partial derivatives on an open region containing the volume V .

Example: Using the Divergence Theorem to find Flux

Use the Divergence Theorem to evaluate the flux integral $\oiint_S \vec{F} \cdot \hat{n} \, dS$, where $\vec{F} = \langle x^2, xy, z \rangle$ and S is the part of the paraboloid $z = 1 - x^2 - y^2$ above the xy -plane, oriented upwards.

First, we need to find the divergence of $\vec{F}$:

$$\operatorname{div} \vec{F} = \frac{\partial x^2}{\partial x} + \frac{\partial xy}{\partial y} + \frac{\partial z}{\partial z} = 2x + x + 1 = 3x + 1.$$

Thus, by the Divergence Theorem:

$$\oiint_S \vec{F} \cdot \hat{n} \, dS = \iiint_V (\operatorname{div} \vec{F}) \, dV = \iiint_V (3x + 1) \, dV.$$

Let's set up the triple integral in cylindrical coordinates. The volume V is bounded by the paraboloid $z = 1 - r^2$ and the xy -plane ($z = 0$), so:

$$\iiint_V (3x + 1) \, dV = \int_0^{2\pi} \int_0^1 \int_0^{1-r^2} (3(r \cos \theta) + 1)r \, dz \, dr \, d\theta.$$

Evaluate the integral:

$$\int_0^{2\pi} \int_0^1 \int_0^{1-r^2} (3(r \cos \theta) + 1)r \, dz \, dr \, d\theta = \frac{\pi}{2}.$$

Example: Using the Divergence Theorem to Evaluate a Triple Integral

Divergence Theorem in Other Dimensions

We can also generalize the divergence theorem to flux integrals in $\mathbb{R}^2$:

$$\oint_C \vec{F} \cdot \hat{n} \, ds = \iint_D (\operatorname{div} \vec{F}) \, dA.$$

Let $\vec{F} = \langle P, Q \rangle$. Then we can alternatively write:

$$\oint_C P \, dy - Q \, dx = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) \, dA$$

This is simply the flux-divergence form of Green's Theorem. In other words, Green's Theorem can also be interpreted as a special case of the Divergence Theorem in two dimensions.

We can also generalize into higher dimensions. For example:

$$\oiint_S \vec{F} \cdot d\vec{V} = \iiint_W (\operatorname{div} \vec{F}) dW.$$

5.9 Generalizing Vector Calculus

Summary of this Unit

1. Line integrals integrate scalar or vector fields over space curves.

- Line integrals over scalar fields sum up little pieces of length $ds = \sqrt{dx^2 + dy^2 + dz^2}$ weighted by the scalar field:

$$\int_C f(\vec{r}) ds = \int_a^b f(\vec{r}(t)) |\vec{r}'(t)| dt.$$

- Line integrals over vector fields sum up little pieces of displacement $d\vec{r} = \langle dx, dy, dz \rangle$ weighted by the component of the vector field aligned with the curve's tangent vector:

$$\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt.$$

2. Surface integrals integrate scalar or vector fields over surfaces.

- Surface integrals over scalar fields sum up little pieces of surface area dS weighted by the scalar field:

$$\iint_S f(\vec{r}(u, v)) dS = \iint_D f(\vec{r}(u, v)) |\vec{r}_u \times \vec{r}_v| du dv.$$

If a surface is defined as $z = g(x, y)$, then:

$$\iint_S f(x, y, z) dS = \iint_D f(x, y, g(x, y)) \sqrt{1 + (g_x)^2 + (g_y)^2} dA.$$

- Flux integrals over vector fields sum up little pieces of surface area dS weighted by the component of the vector field aligned with the surface's normal vector:

$$\iint_S \vec{F} \cdot \hat{n} dS = \iint_D \vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) du dv.$$

3. Four Big Theorems of Vector Calculus:

- Fundamental Theorem of Calculus for Line Integrals (Gradient Theorem)**

Sum of small changes of a scalar field over a curve equals the net change over the endpoints.

$$\int_C \vec{\nabla} f \cdot d\vec{r} = f(\vec{q}) - f(\vec{p})$$

- Green's Theorem**

Generalization of Stokes' and Divergence Theorems to 2D.

$$\oint_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA \quad (\text{circulation-curl form})$$

$$\oint_C P dy - Q dx = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dA \quad (\text{flux-divergence form})$$

- Stokes' Theorem (Curl Theorem)**

Sum of little curls enclosed by a curve equals the circulation around the curve.

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_D (\text{curl } \vec{F}) \cdot \hat{n} dS$$

- **Divergence Theorem (Gauss' Theorem)**

Sum of little divergences inside a volume equals the flux through the volume's boundary.

$$\oiint_S \vec{F} \cdot \hat{n} dS = \iiint_V (\text{div } \vec{F}) dV$$

Notice that all four theorems follow the same general pattern of relating an integral over a boundary to an integral (or net change) over the region enclosed by that boundary.

Generalizing the Theorems

Generalized Stokes' Theorem

Let S be some n -dimensional orientable manifold and ∂S be the boundary of that manifold in $n - 1$ dimensions. Then:

$$\int_S d\omega = \int_{\partial S} \omega.$$

This theorem generalizes the general theme of the fundamental theorems of calculus, which always go something along the lines of “the integral of some differential (“sum of little changes”) over some boundary is equal to the total change over the boundary.

Differential Forms

Differential k-Form

A *differential k-form* ω is an expression that can be integrated over a k -dimensional manifold. Examples include:

- A differential 1-form in $\mathbb{R}^2$: $\omega = P dx + Q dy$
- A differential 2-form in $\mathbb{R}^3$: $\omega = P dy dz + Q dz dx + R dx dy$

Exterior Derivative

The *exterior derivative* $d\omega$ of a differential form ω is a new differential form that represents the infinitesimal change of ω . Examples include:

- If $\omega = P dx + Q dy$ is a differential 1-form in $\mathbb{R}^2$, then:

$$d\omega = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy.$$

- If $\omega = P dy dz + Q dz dx + R dx dy$ is a differential 2-form in $\mathbb{R}^3$, then:

$$d\omega = \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dx dy dz.$$

Heimholtz Decomposition

We can decompose any vector field into a sum of a zero-curl vector field $\vec{G}$ and a zero-divergence vector field $\vec{R}$:

$$\vec{F} = \vec{G} + \vec{R}.$$

- The zero-curl vector field $\vec{G}$ is called the **irrotational** component of $\vec{F}$. It is called $\vec{G}$ because it has curl zero, meaning it is conservative and can be written as the gradient of some scalar potential function f : $\vec{G} = \vec{\nabla}f$.
- The zero-divergence vector field $\vec{R}$ is called the **solenoidal** component of $\vec{F}$.

Heimholtz Decomposition Theorem

Let $\vec{F}$ be some differentiable vector field on $\mathbb{R}^3$. Then there must exist a potential function φ and a vector potential $\vec{A}$ such that:

$$\vec{F} = -\vec{\nabla}\varphi + \text{curl } \vec{A}.$$

It follows that:

- $-\vec{\nabla}\varphi$ is the irrotational (zero-curl) component of $\vec{F}$.
- $\text{curl } \vec{A}$ is the solenoidal (zero-divergence) component of $\vec{F}$.